

Palpation-Assisted
Tomography
(PAT)
Scan

PhD COMPREHENSIVE EXAM Presentation

Vivek K

THE DOCTOR'S HANDS

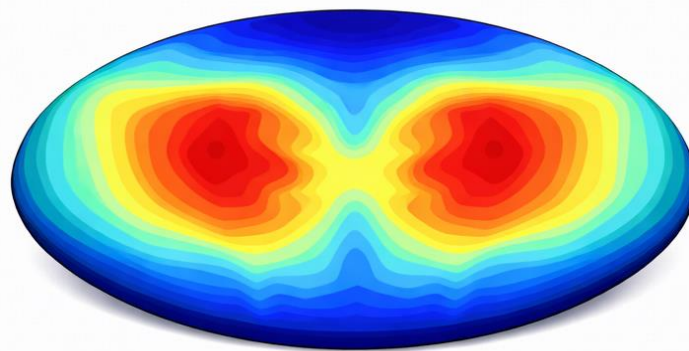


THE CORE PROBLEM

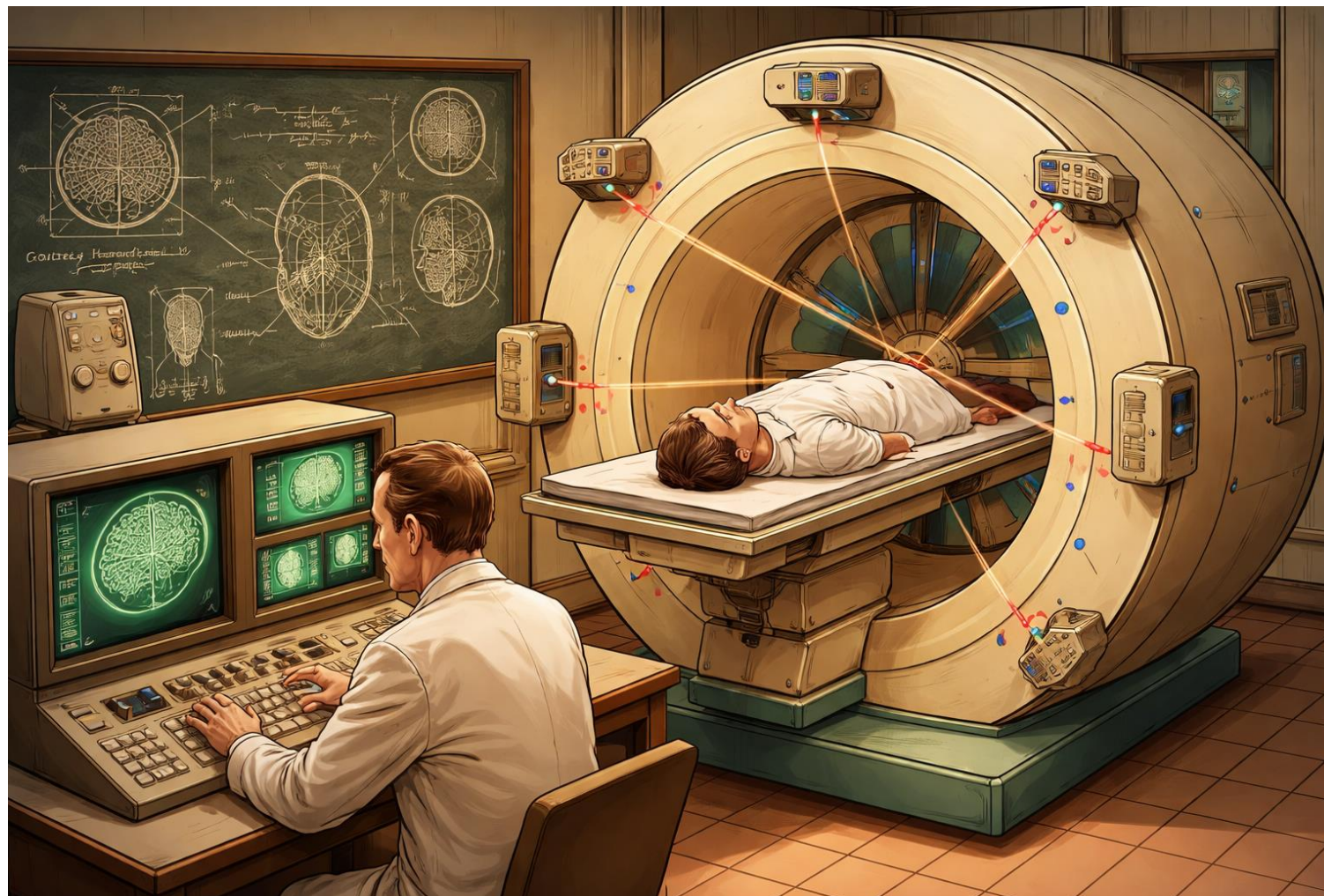


What's inside?
Can you guess?

BOUNDARY MEASUREMENTS ONLY

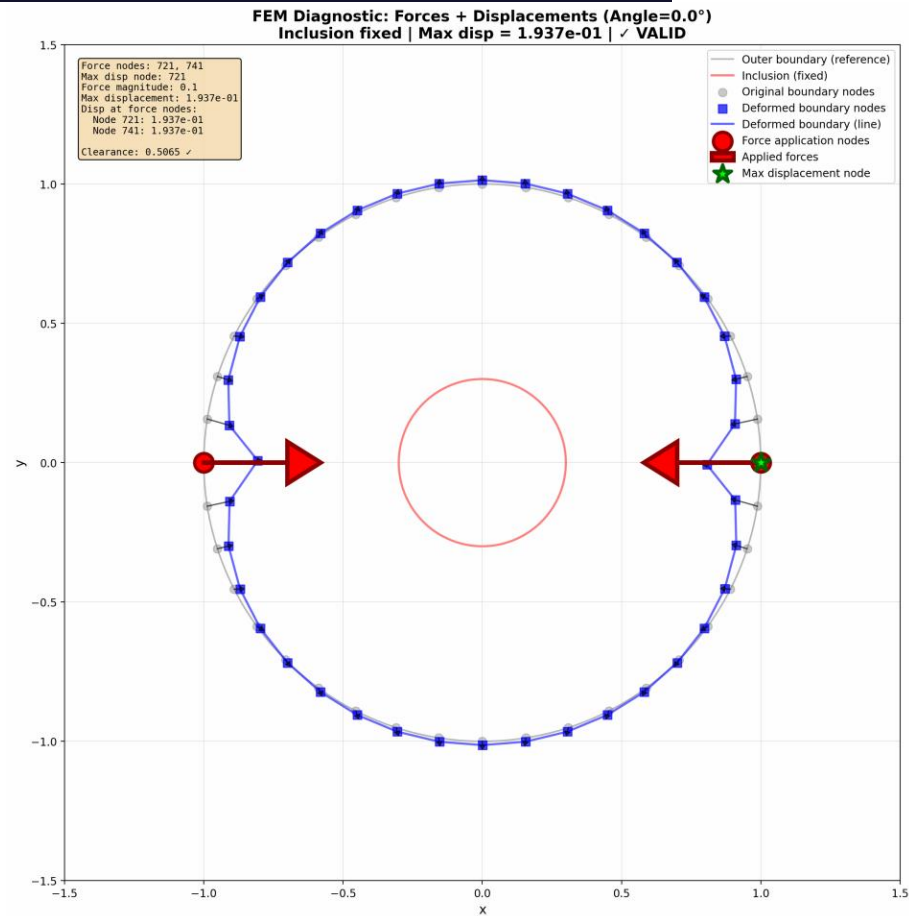


CT SCANS



LEARNING FROM CT SCANS

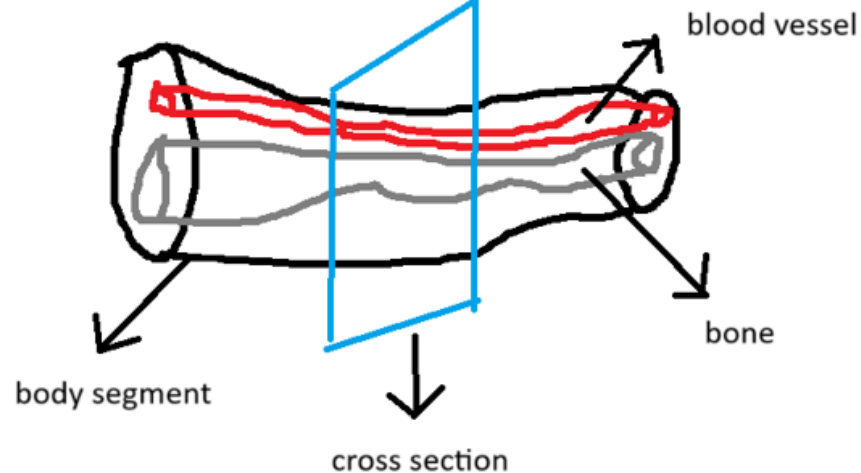
"What if we could do for touch what CT did for X-rays?"



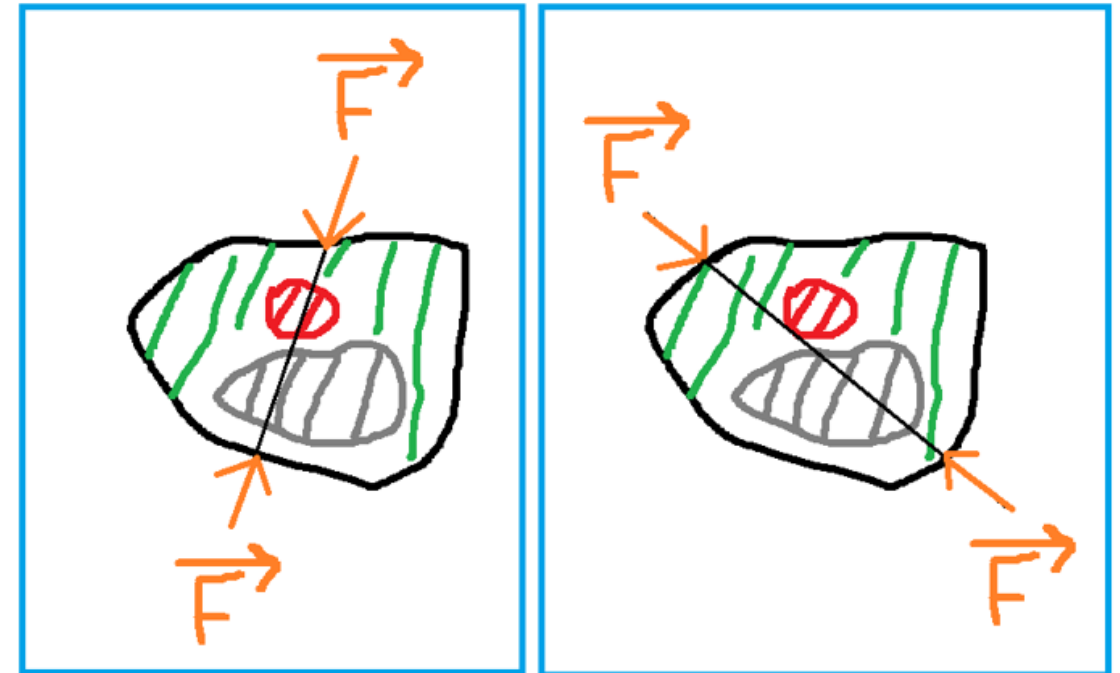
Palpation-Assisted Tomography (PAT) Scan

- Known loading condition over small area
- Surface Deformation available

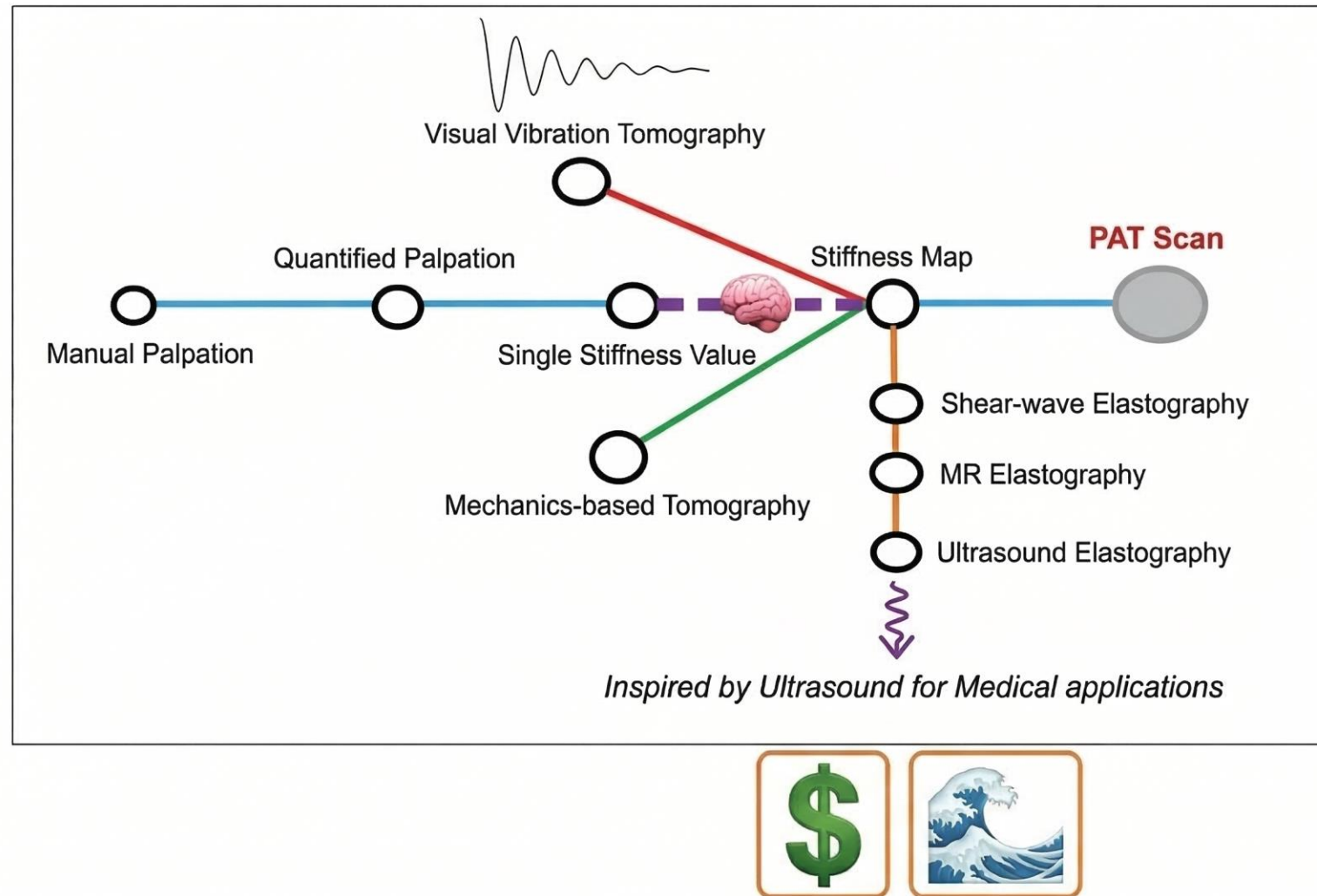
- 2D cross sections
- Span the cross-section by applying equal and opposite radial forces
- Analogous picture to Tomographic reconstruction



cross sectional views



BUILDING THE SCIENTIFIC LANDSCAPE AS "THE PAT SCAN SUBWAY"



STANDING ON THE SHOULDERS OF GIANTS...



sensors



Article

Mechanics Based Tomography: A Preliminary Feasibility Study

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² Department of Mathematics, Texas A&M University, College Station, TX 77843, USA; sichengwang0223@gmail.com

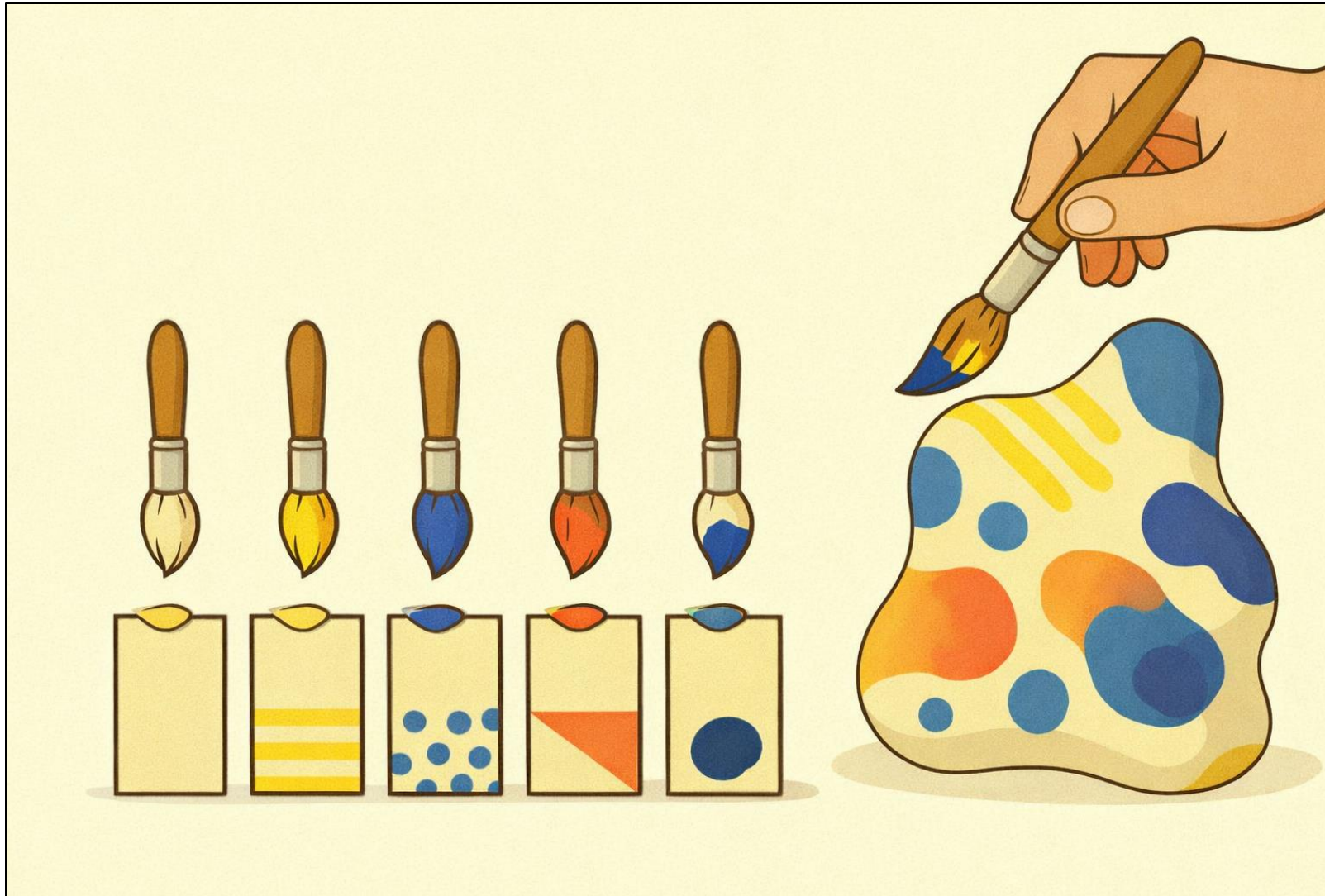
* Correspondence: sevangoenezen@gmail.com; Tel.: +1-979-862-2786

Academic Editor: Iulian I. Iordachita

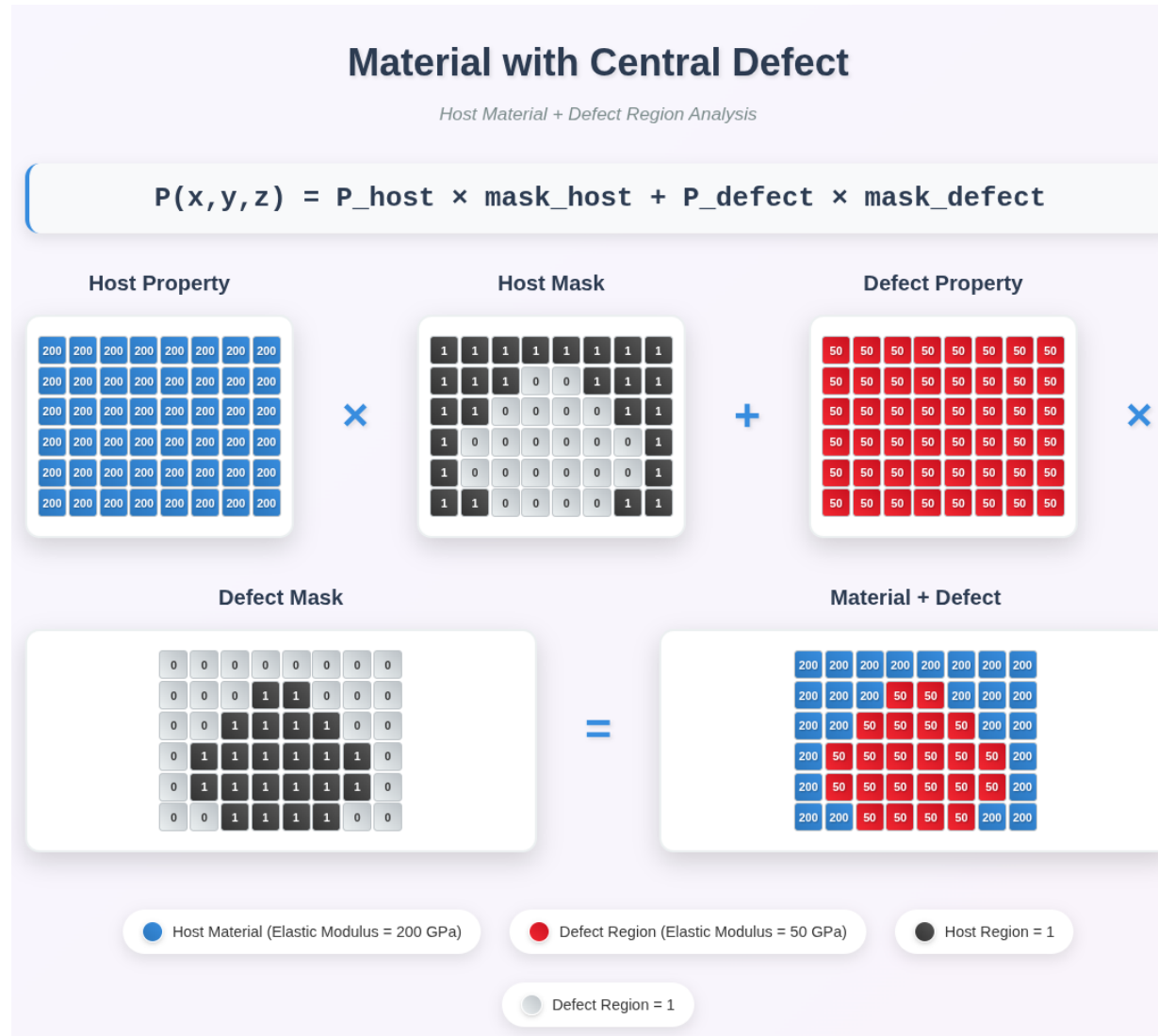
Received: 28 February 2017; Accepted: 3 May 2017; Published: 9 May 2017

Abstract: We present a non-destructive approach to sense inclusion objects embedded in a solid medium remotely from force sensors applied to the medium and boundary displacements that could be measured via a digital image correlation system using a set of cameras. We provide a rationale and strategy to uniquely identify the heterogeneous sample composition based on stiffness (here, shear modulus) maps. The feasibility of this inversion scheme is tested with simulated experiments that could have clinical relevance in diagnostic imaging (e.g., tumor detection) or could be applied to engineering materials. No assumptions are made on the shape or stiffness quantity of the inclusions. We observe that the novel inversion method using solely boundary displacements and force measurements performs well in recovering the heterogeneous material/tissue composition that consists of one and two stiff inclusions embedded in a softer background material. Furthermore, the target shear modulus value for the stiffer inclusion region is underestimated and the inclusion size is overestimated when incomplete boundary displacements on some part of the boundary are utilized. For displacements measured on the entire boundary, the shear modulus reconstruction improves significantly. Additionally, we observe that with increasing number of displacement data sets utilized in solving the inverse problem, the quality of the mapped shear moduli improves. We also analyze the sensitivity of the shear modulus maps on the noise level varied between 0.1% and 5% white Gaussian noise in the boundary displacements, force and corresponding displacement indentation. Finally, a sensitivity analysis of the recovered shear moduli to the depth, stiffness and the shape of the stiff inclusion is performed. We conclude that this approach has potential as a novel imaging modality and refer to it as Mechanics Based Tomography (MBT).

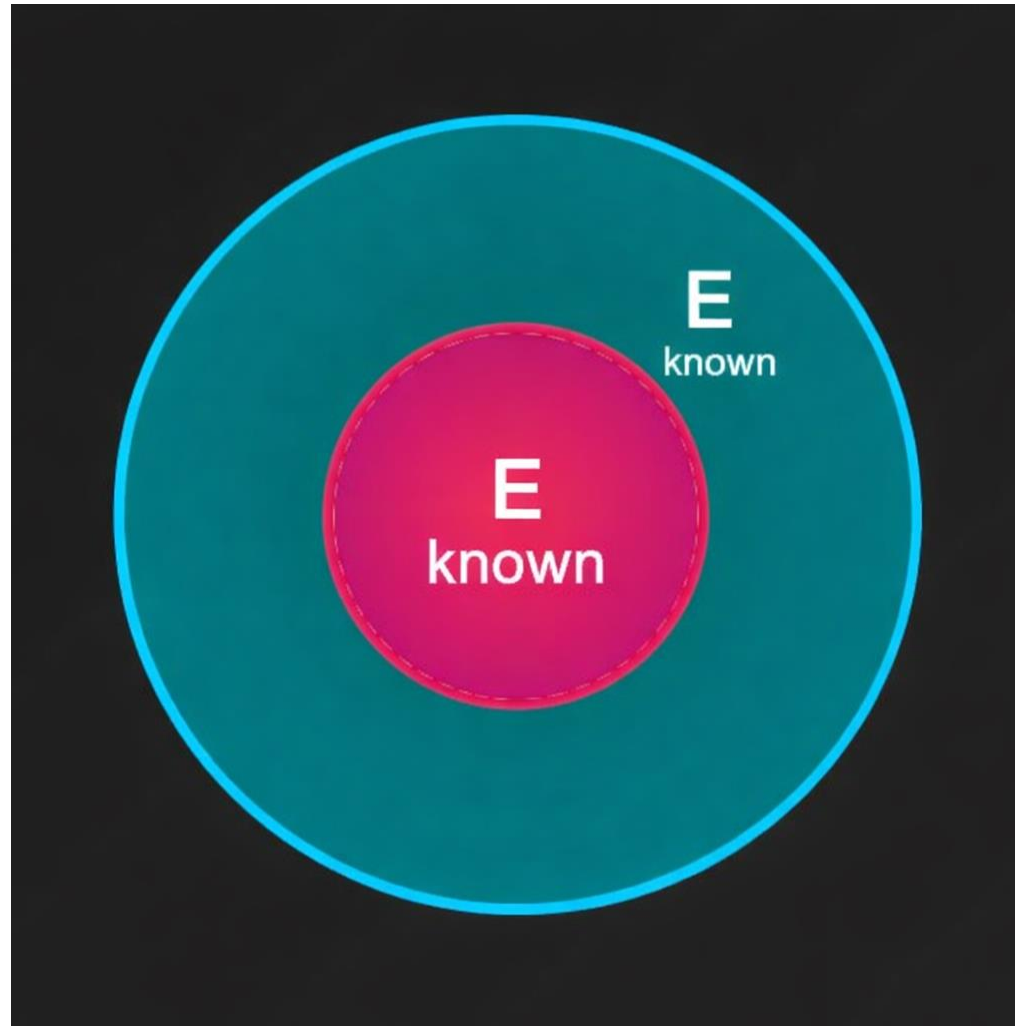
THE PAINTBRUSH REVEALS...



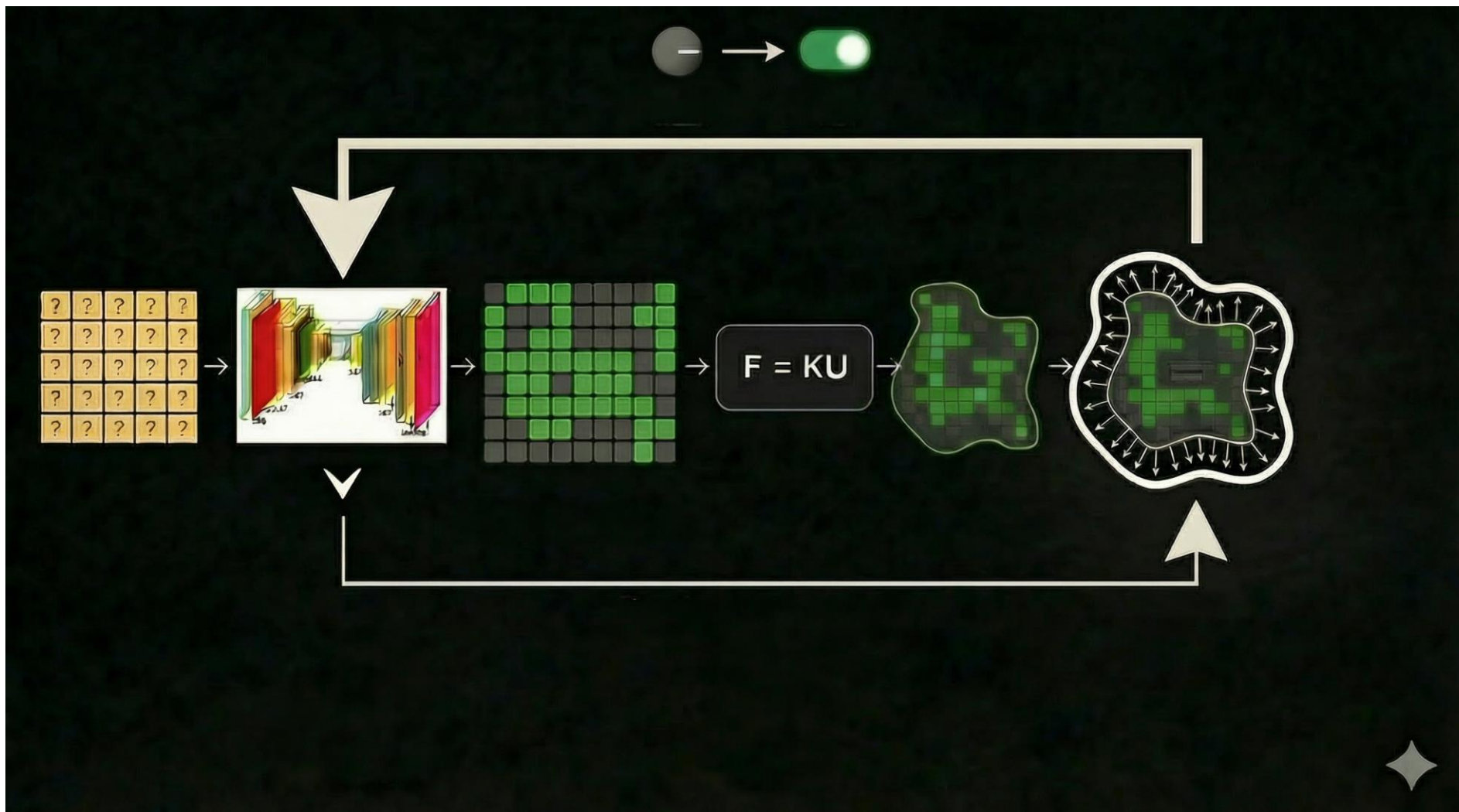
THE PAINTBRUSH INSPIRES...



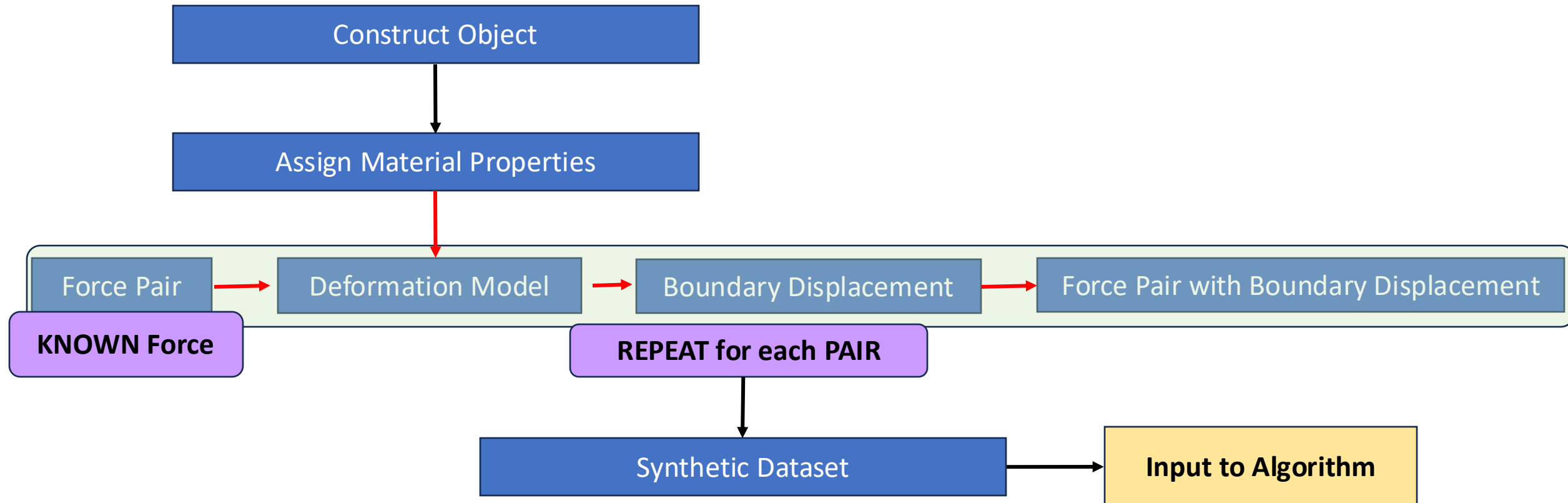
BREAKING DOWN THE PROBLEM



THE CORE IDEA



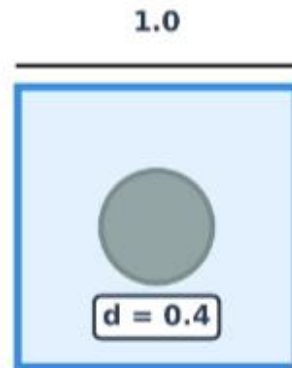
SYNTHETIC DATASET GENERATION: INITIAL IDEA



CONSTRUCTING THE OBJECT

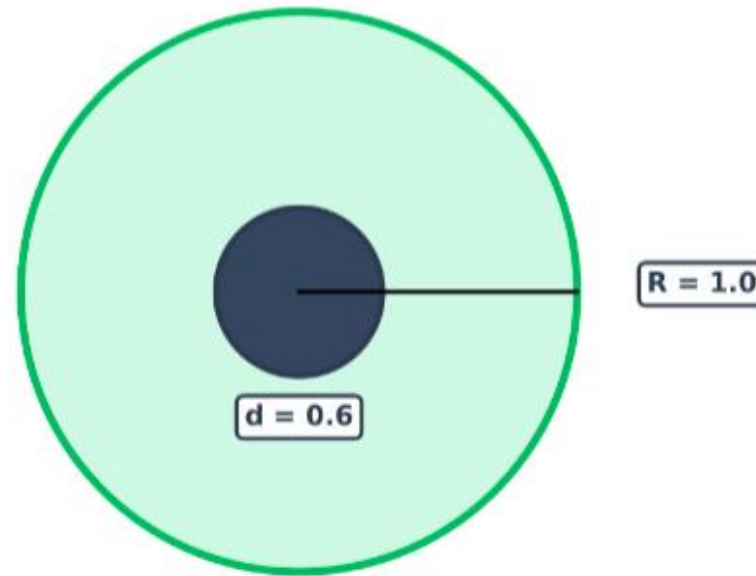
Based on MBT methods paper

MBT



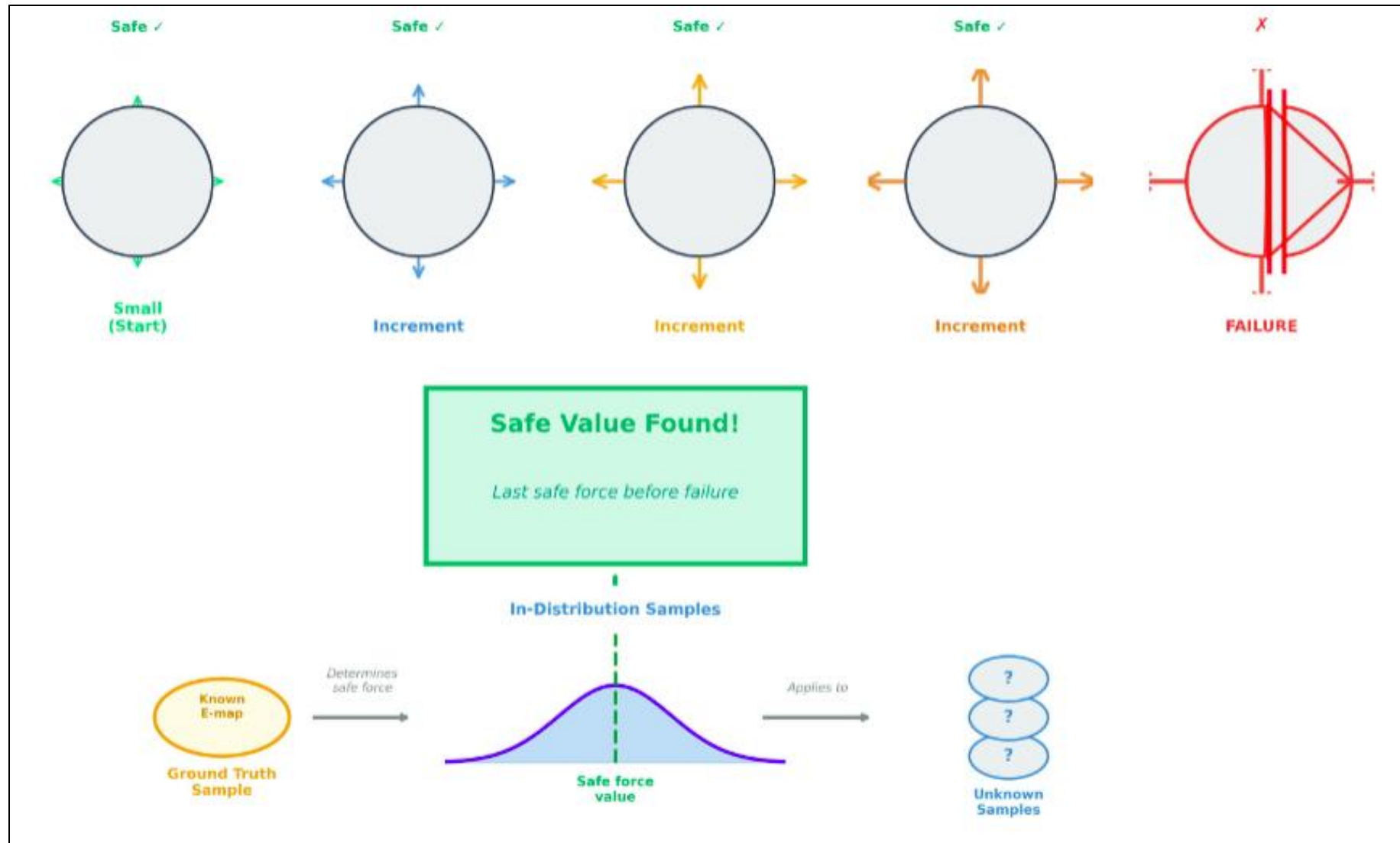
Stiffness Contrast
5×

PAT Scan (Ours)



Stiffness Contrast
10×

DETERMINING THE FORCE MAGNITUDE



DETERMINING THE ANGULAR RESOLUTION



Angular spacing between measurements

Computational Criterion



Uses in-distribution samples

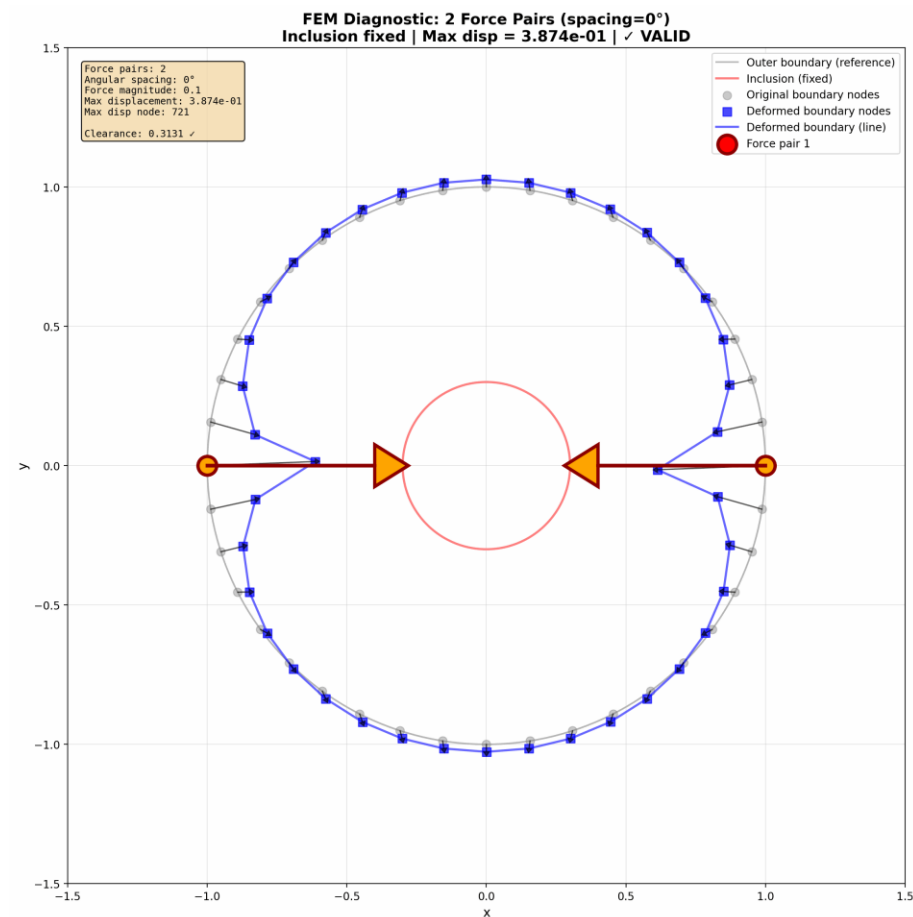
Analytical Criterion



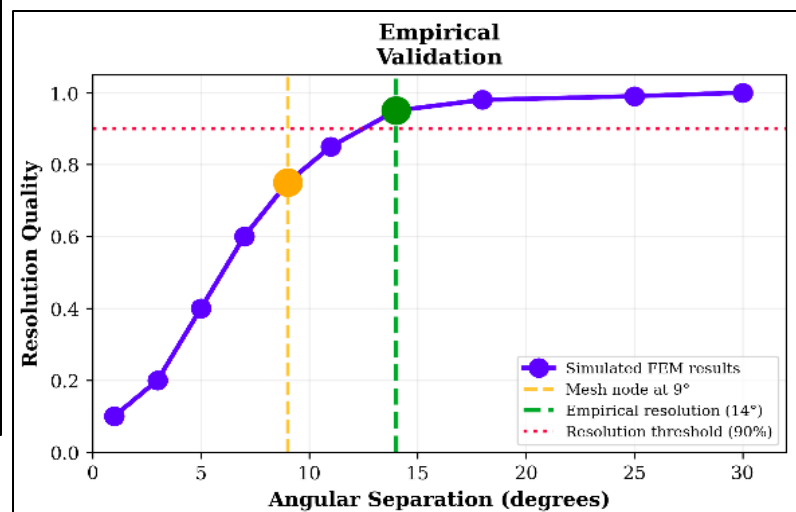
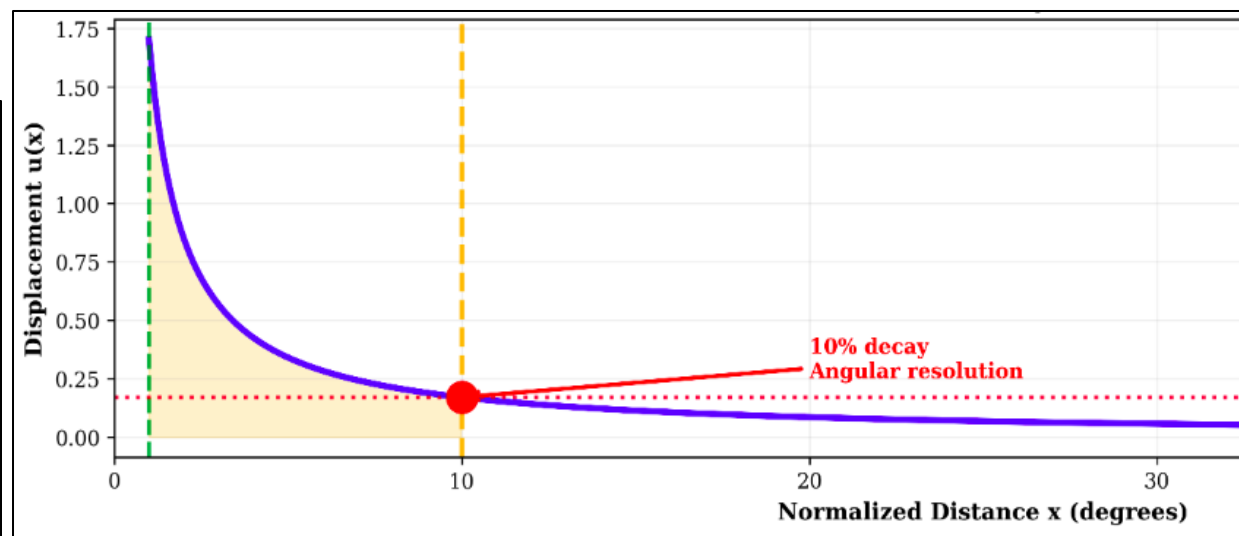
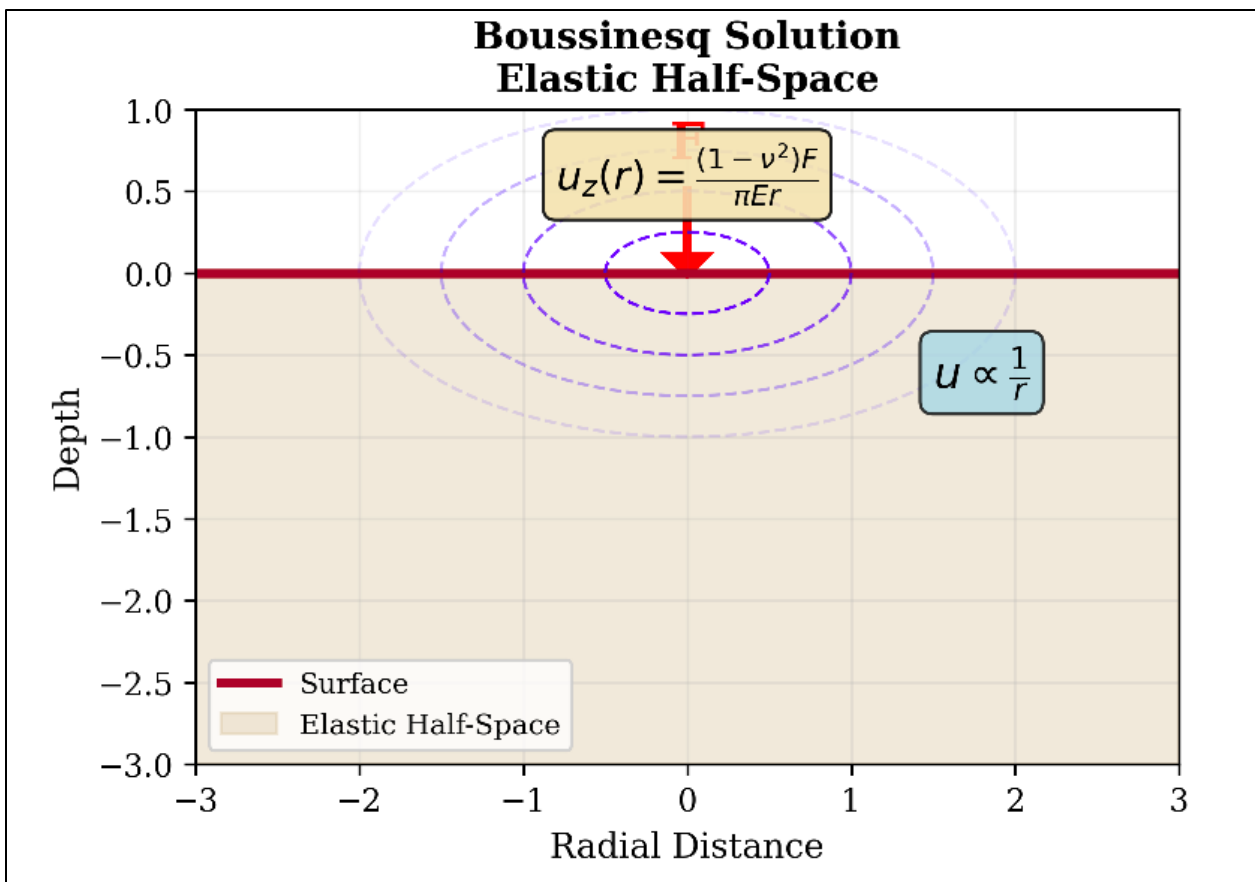
Details in upcoming slides

Two approaches to determine angular resolution

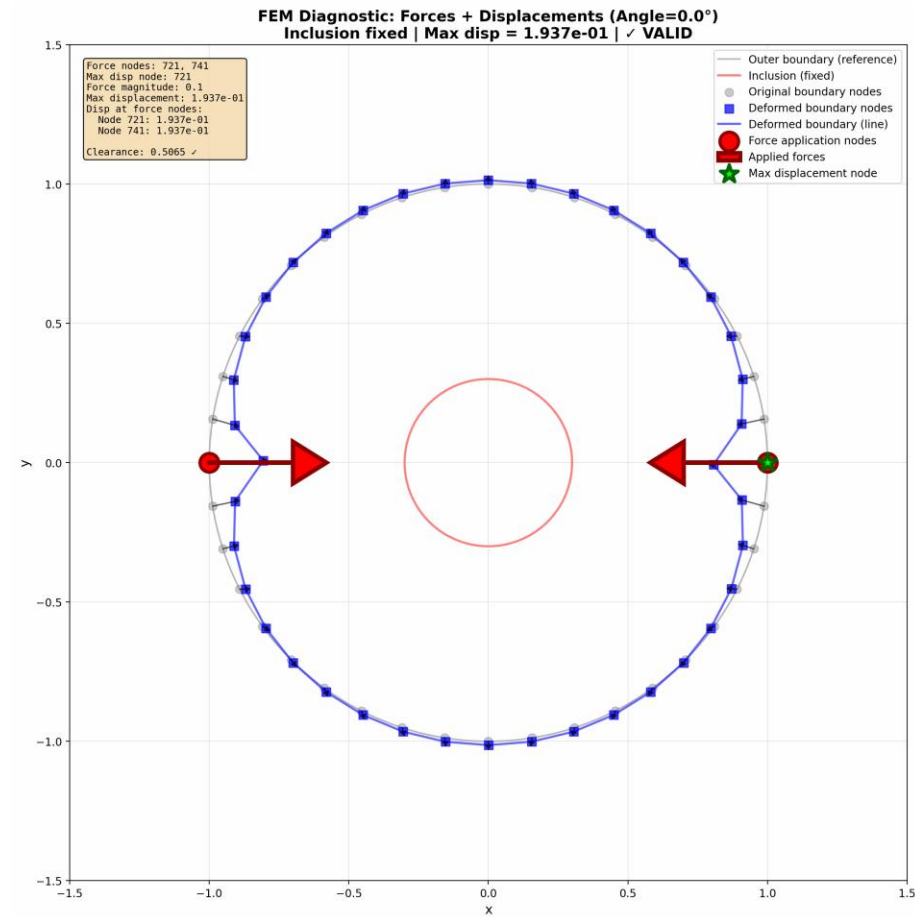
COMPUTATIONAL CRITERION FOR ANGULAR RESOLUTION



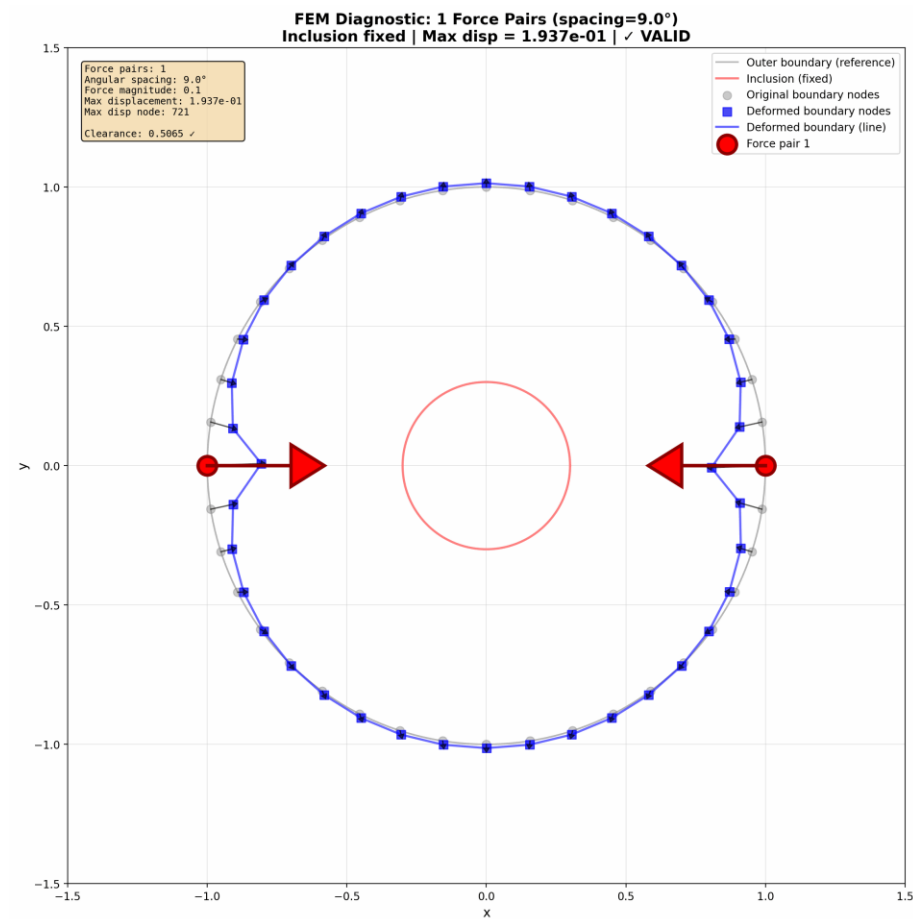
ANALYTICAL CRITERION FOR ANGULAR RESOLUTION



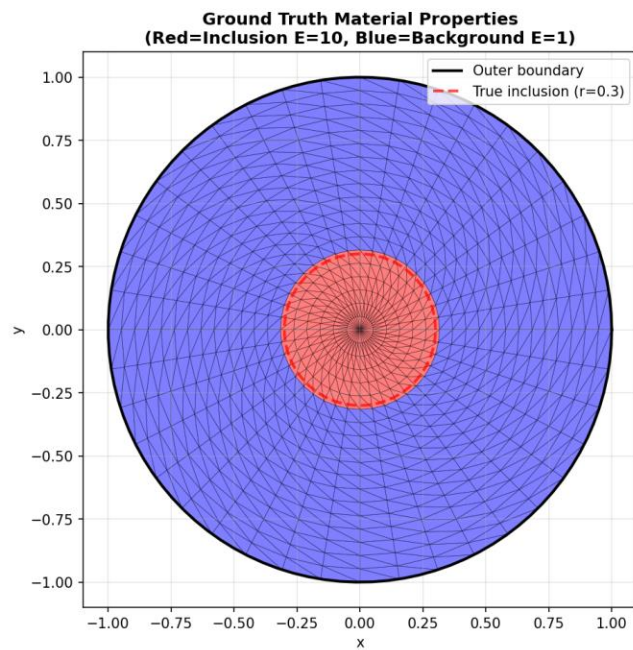
SYMMETRIC SCANNING DATA COLLECTION



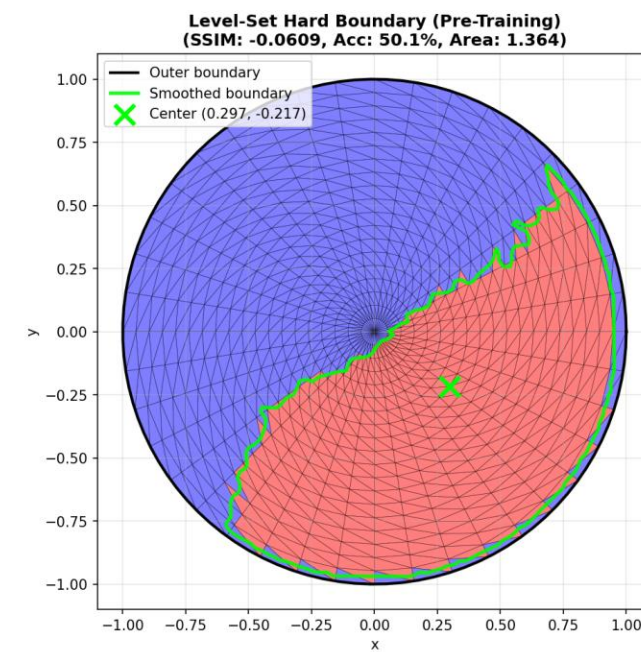
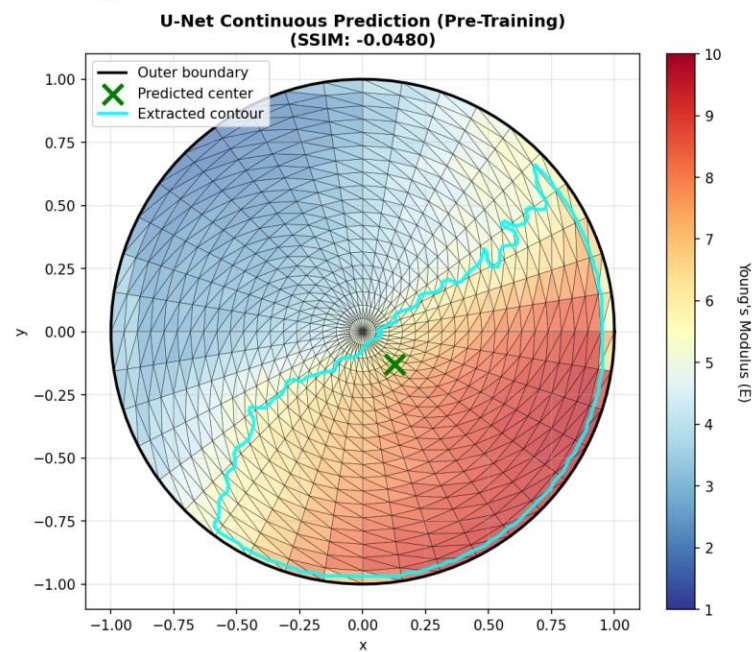
ANGULAR SCANNING DATA COLLECTION



BEFORE TRAINING THE AI

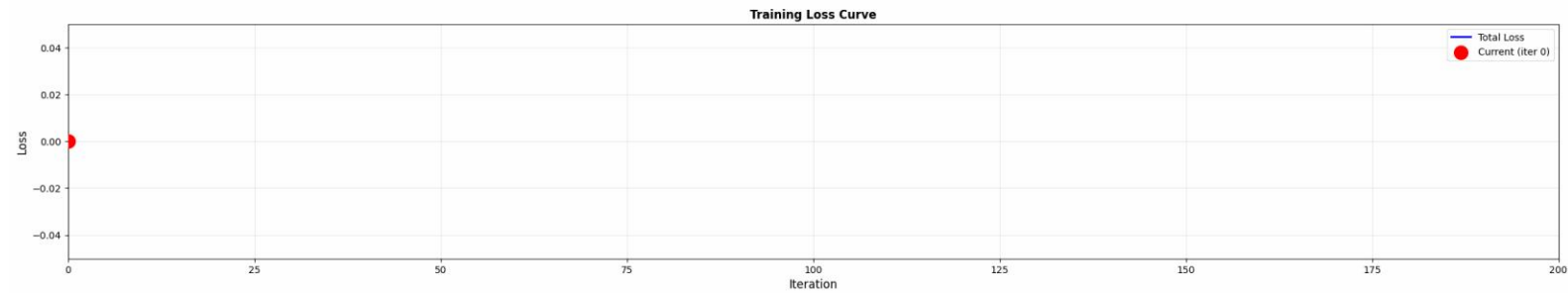
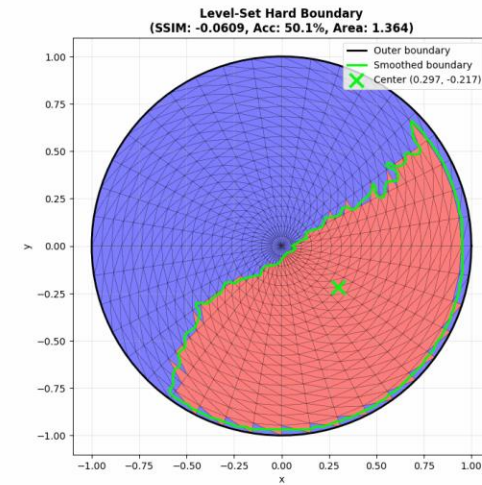
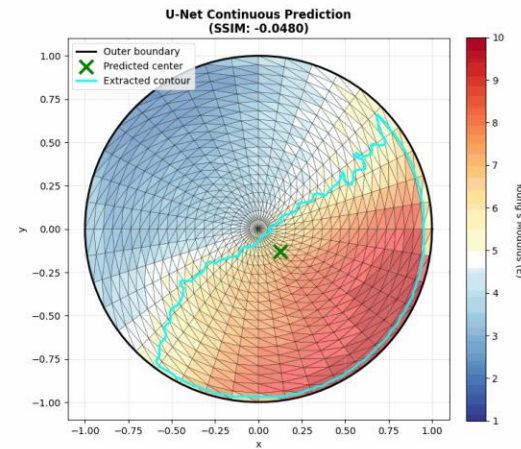
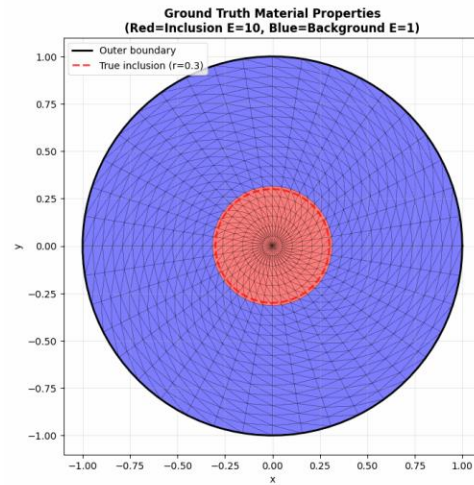


Pre-Training Results (Iteration 0) - Random Initialization

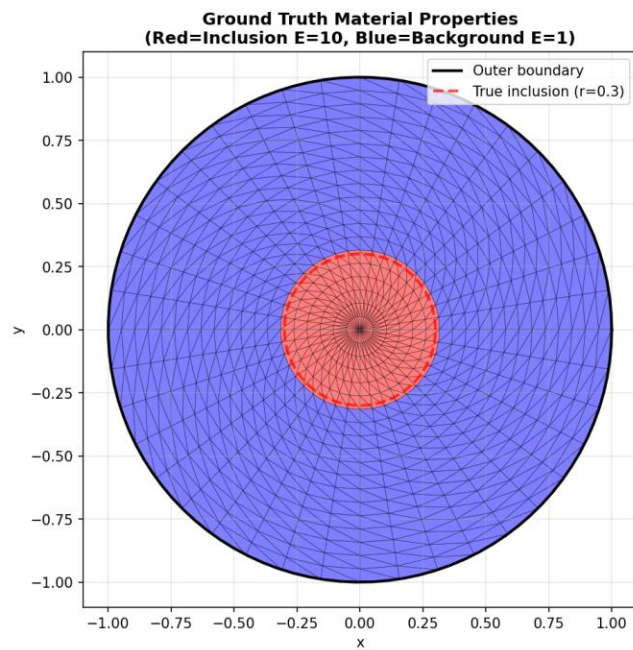


TRAINING THE AI... (SHOW GIF)

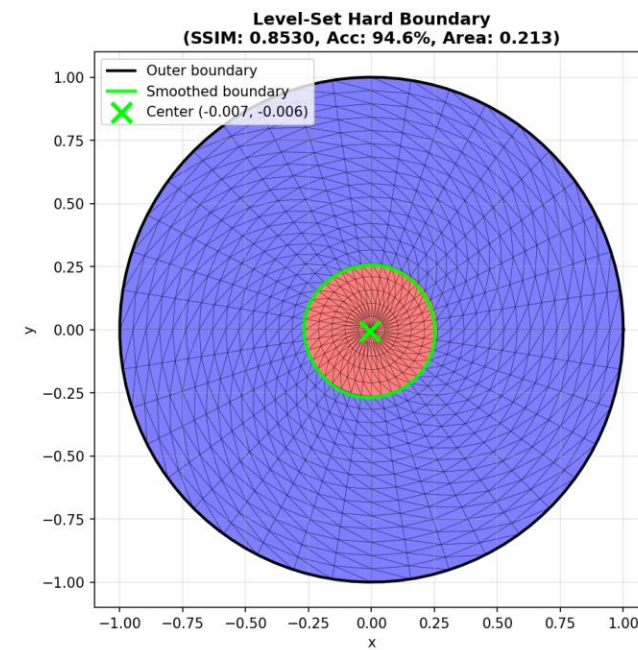
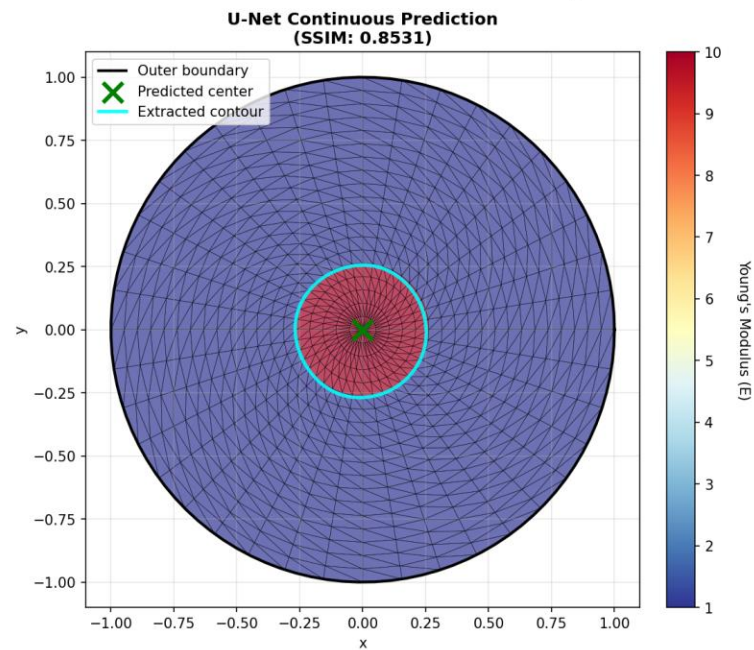
Iteration: 0/200 | Loss: 0.000000 | Time: 00:00



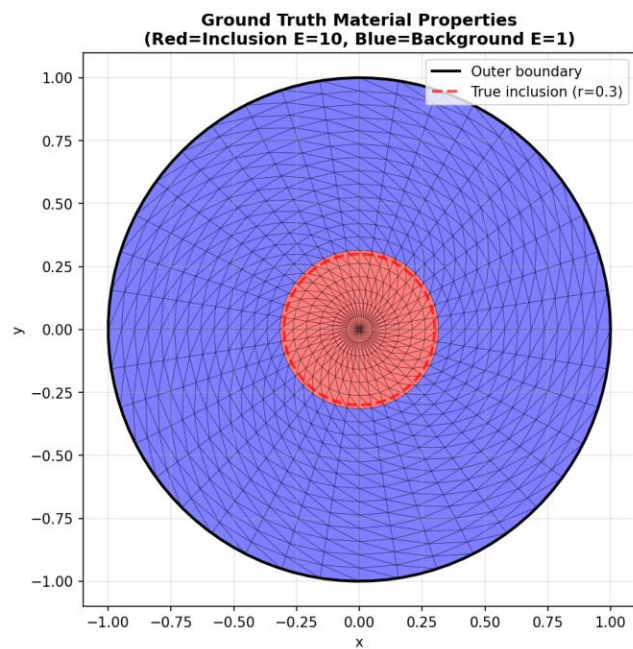
AFTER TRAINING THE AI...



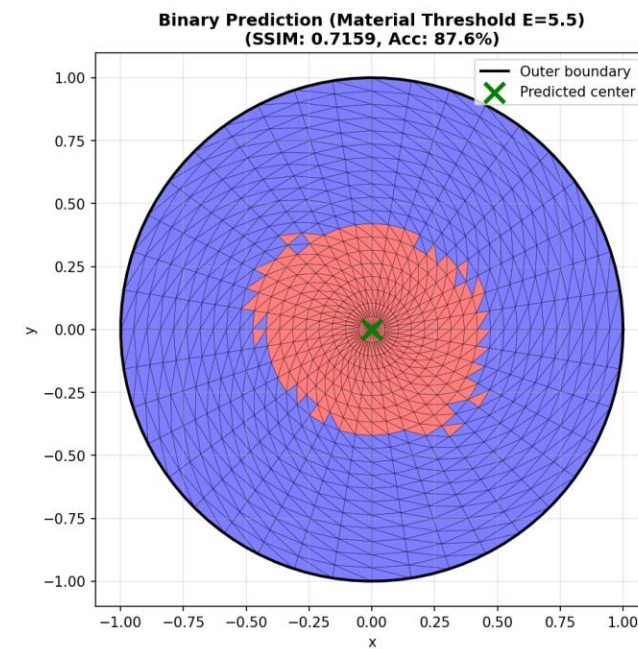
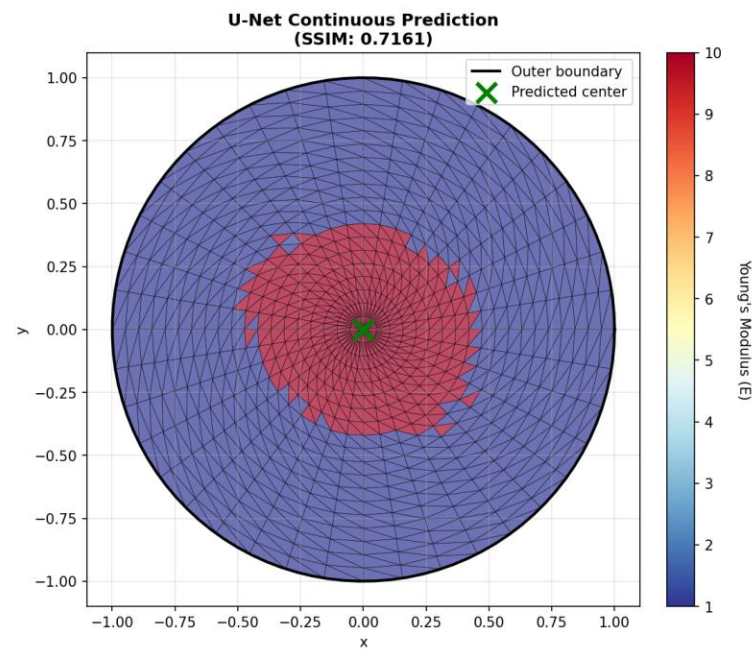
Final Results After 200 Iterations (Grid Search Optimized)



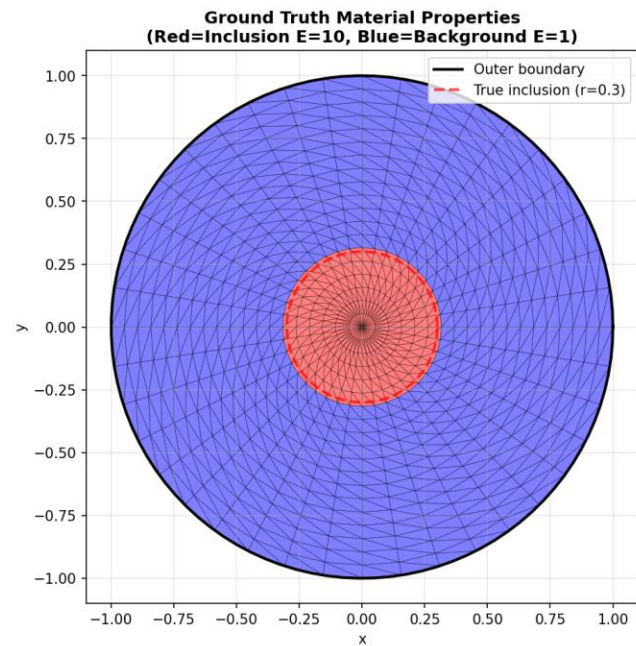
VANILLA RESULTS



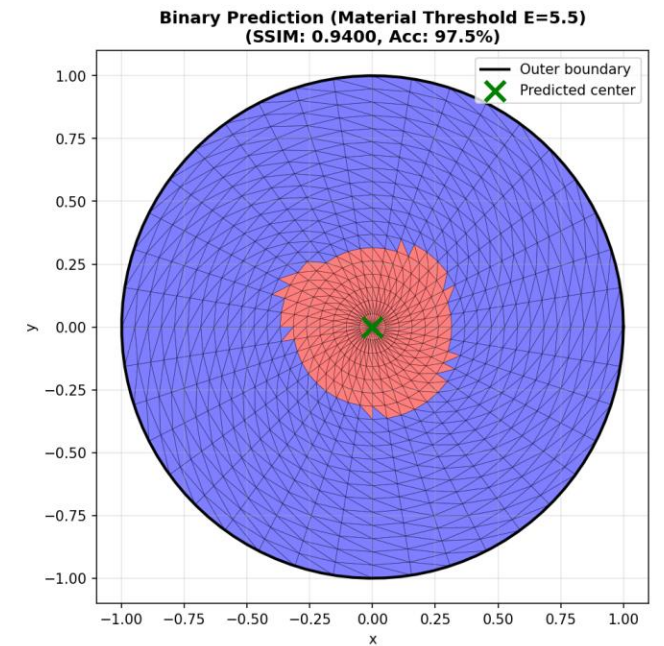
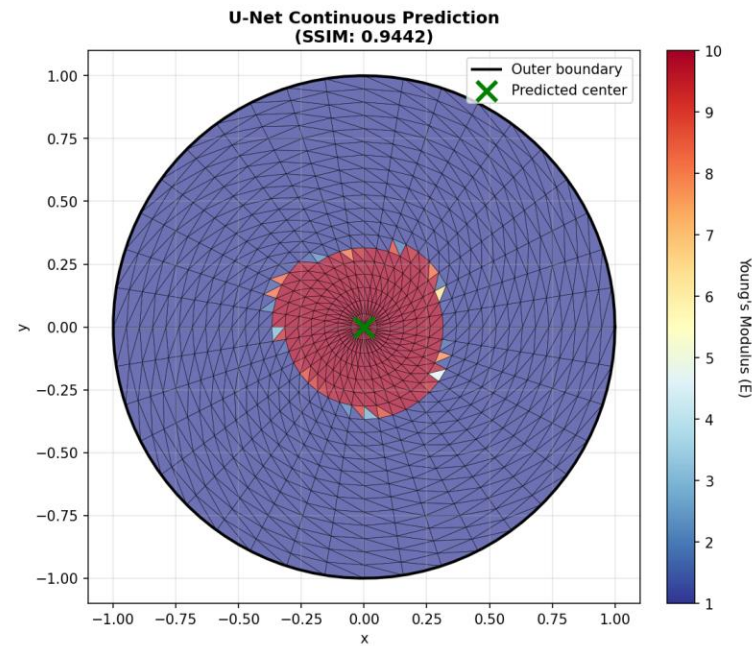
Final Results After 1000 Iterations ($\sqrt{L2}$ Loss)



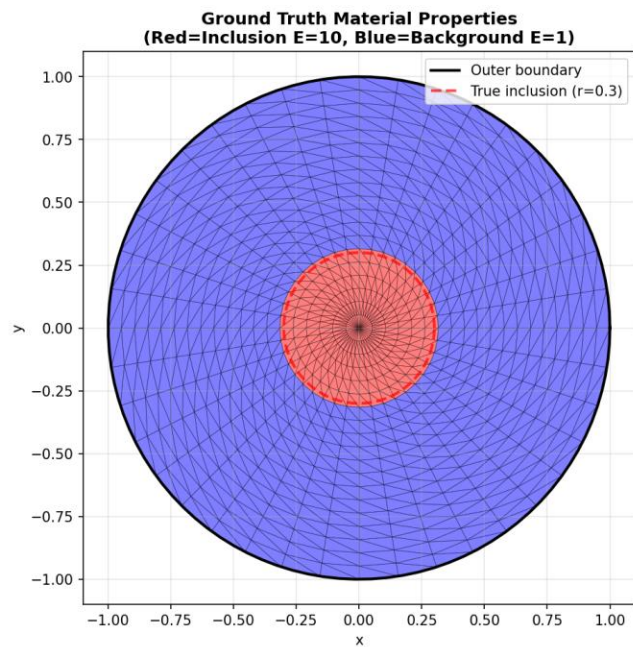
TV REGULARIZATION



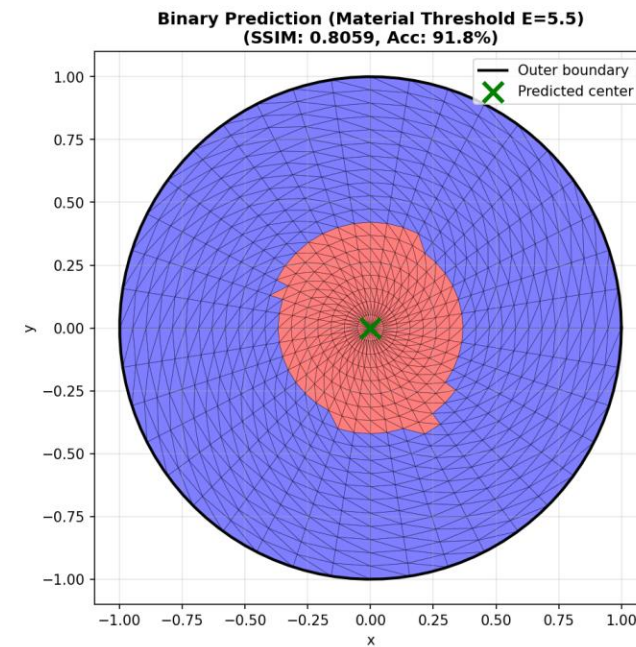
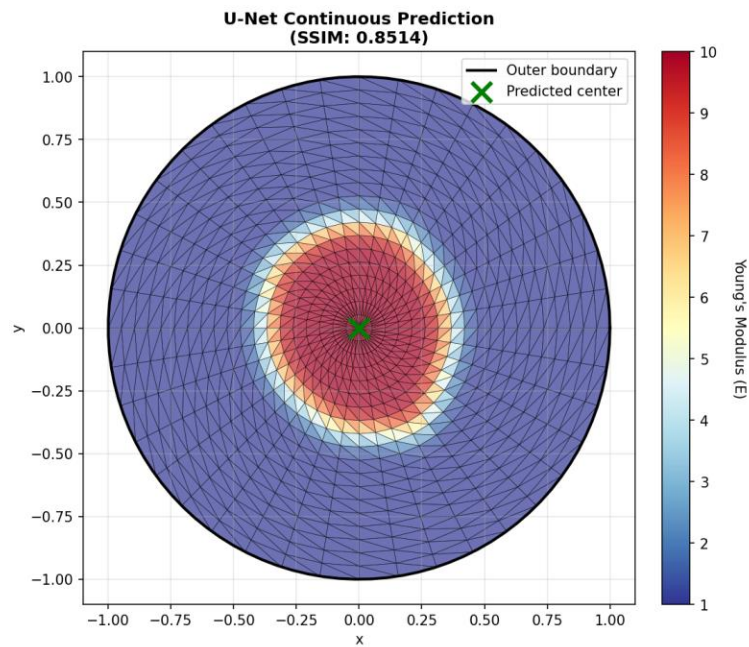
Final Results After 1000 Iterations ($\sqrt{L2}$ Loss + TV Regularization)



TV REGULARIZATION + SMOOTHING

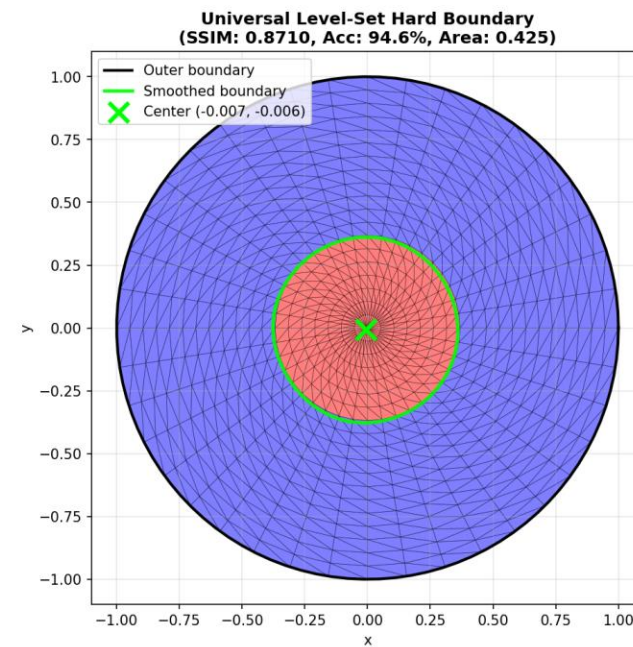
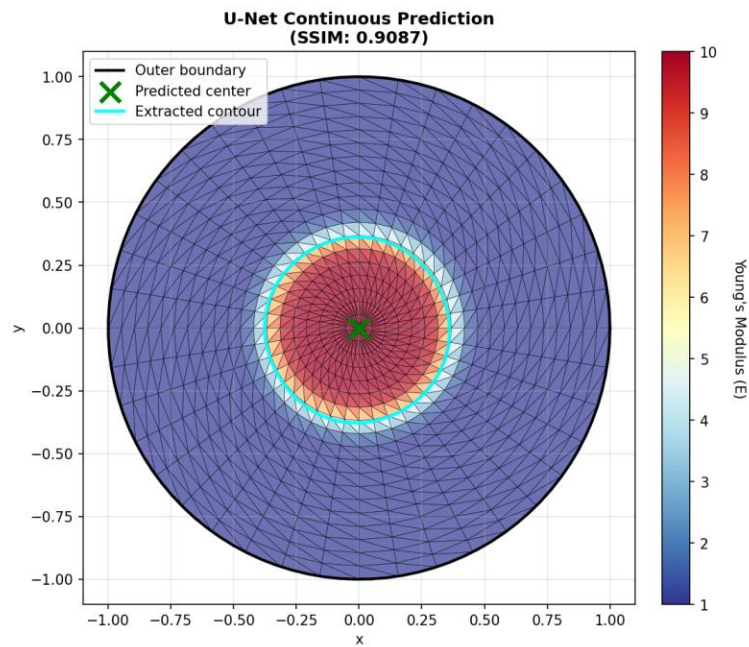
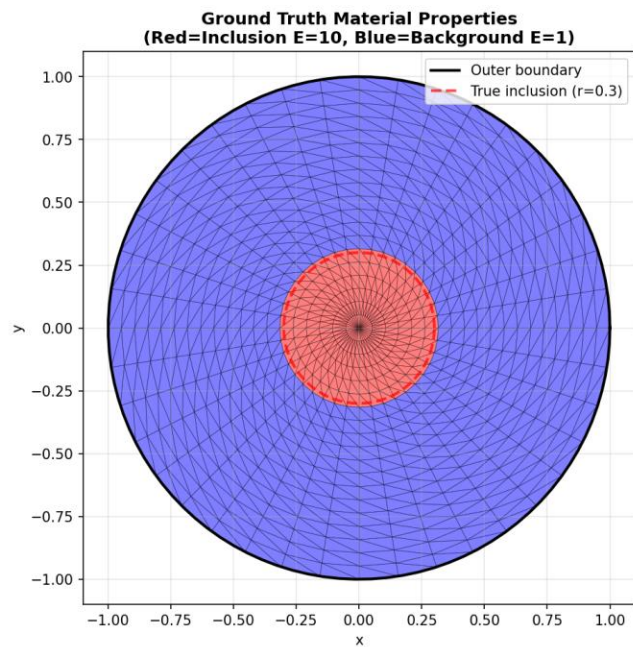


Final Results After 1000 Iterations ($\sqrt{L2}$ Loss + TV Reg + Smoothing)

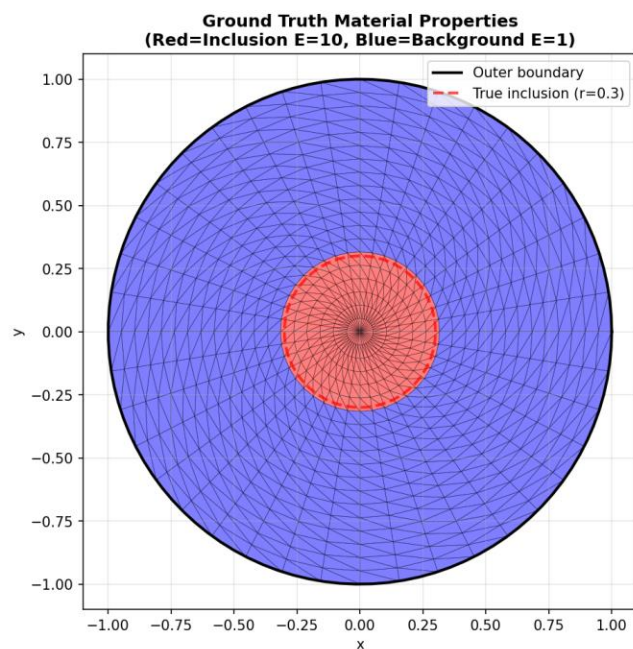


TV REGULARIZATION + SMOOTHING + LEVEL-SET POST-PROCESSING

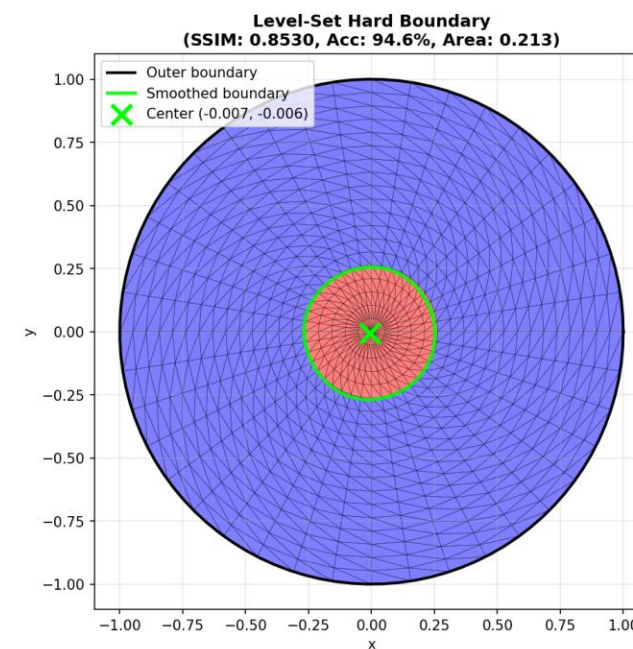
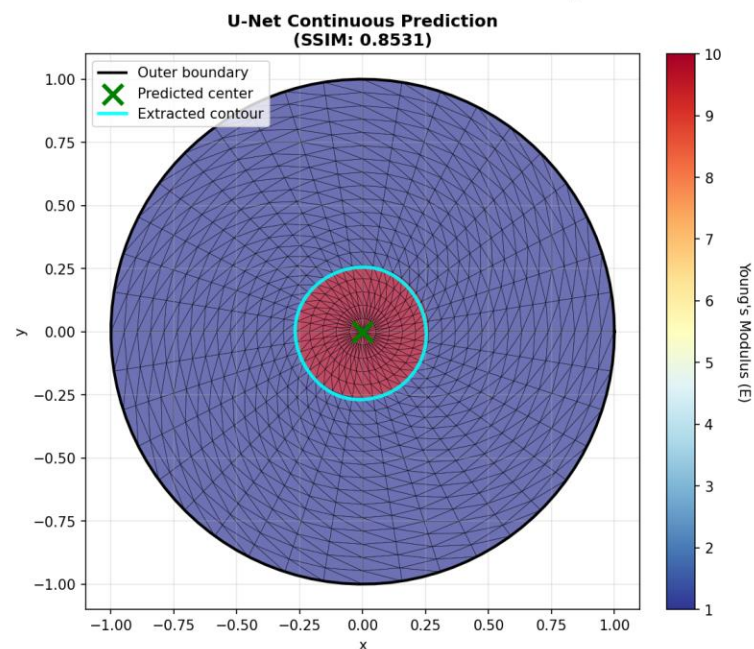
Final Results After 1000 Iterations ($\sqrt{L2}$ + TV + Smoothing + Simple Level-Set)



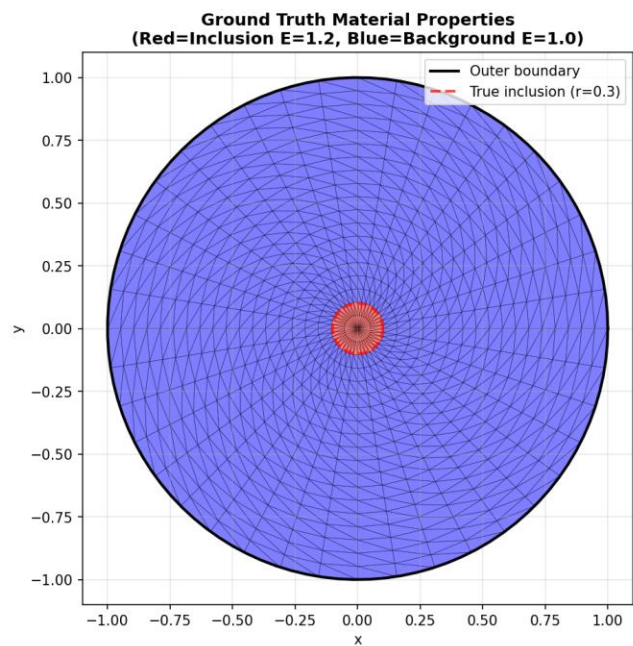
TV REGULARIZATION + SMOOTHING + LEVEL-SET PP + GRID SEARCH



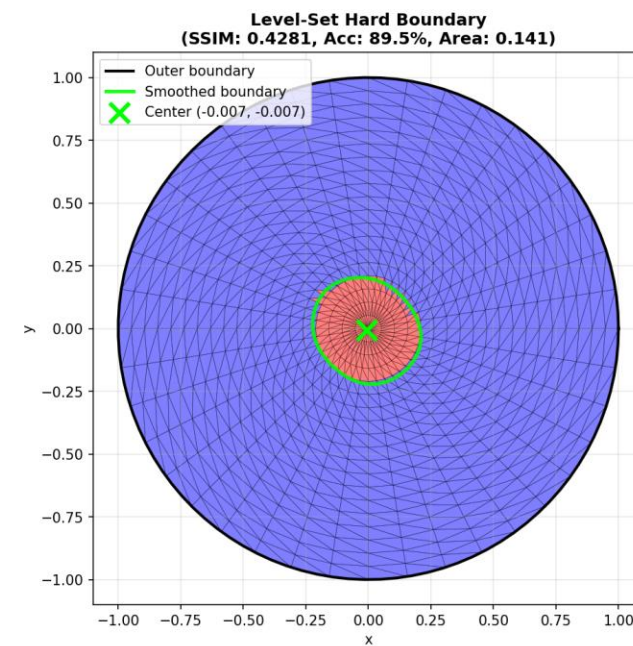
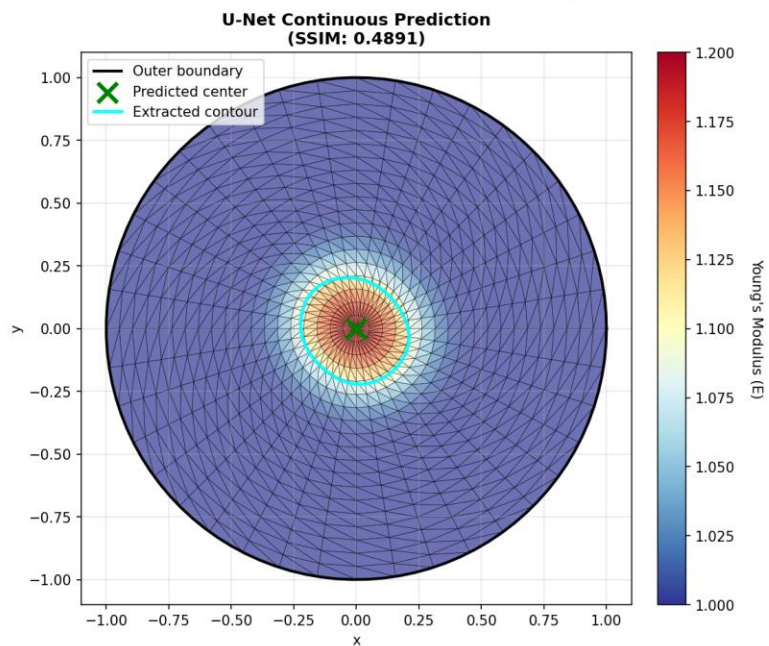
Final Results After 200 Iterations (Grid Search Optimized)



CIRCULAR INCLUSION EDGE CASE

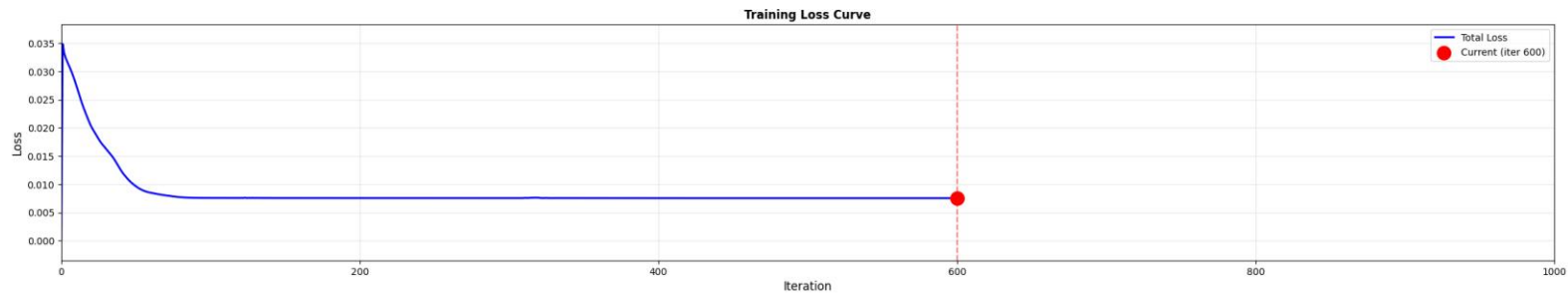
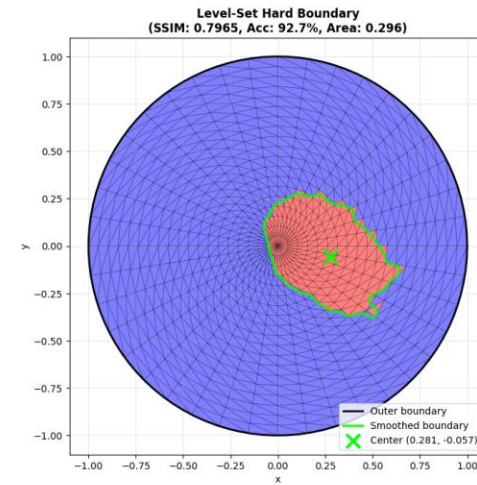
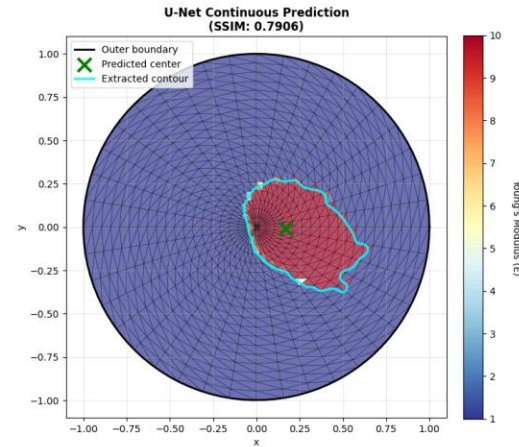
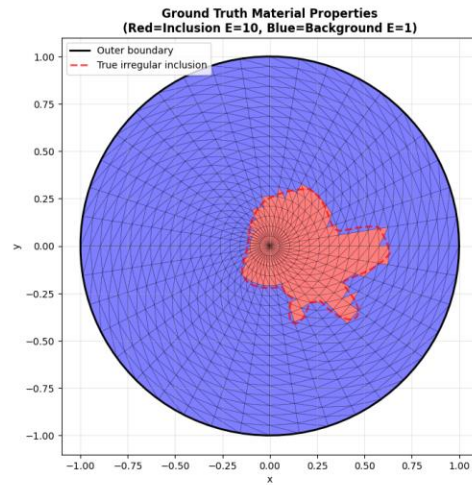


Final Results After 200 Iterations (Grid Search Optimized)

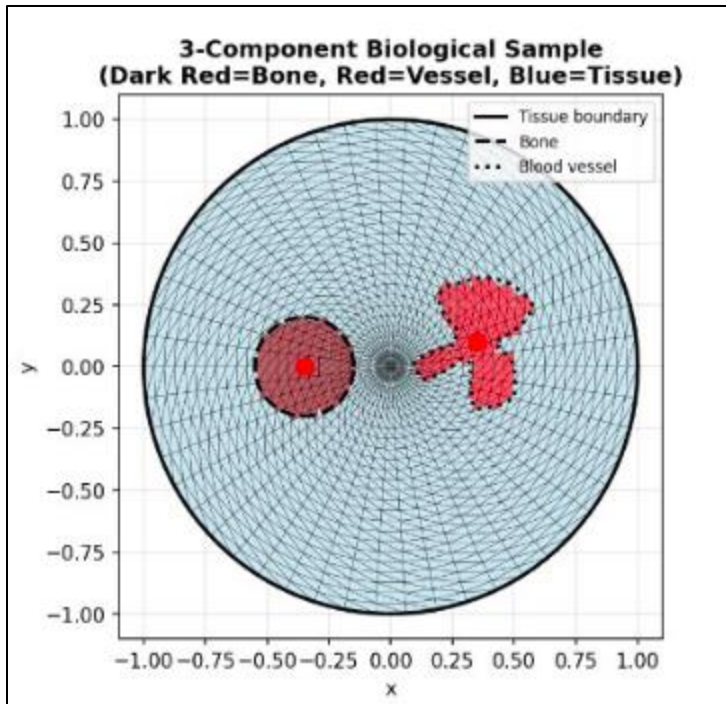


IRREGULAR INCLUSION

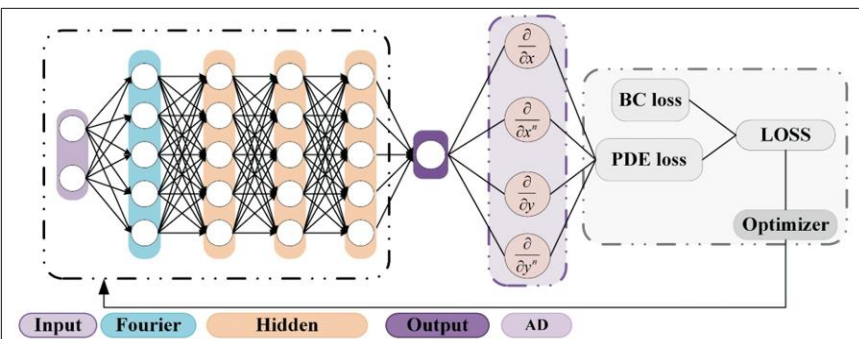
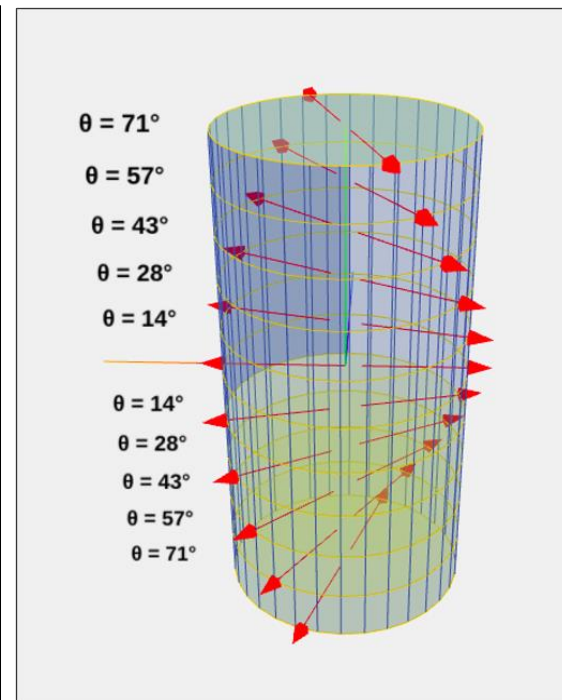
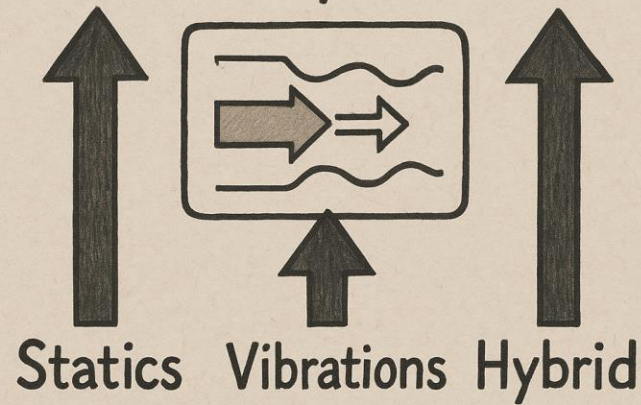
Iteration: 600/1000 | Loss: 0.007529 | Time: 16:33 | Geometry: Irregular



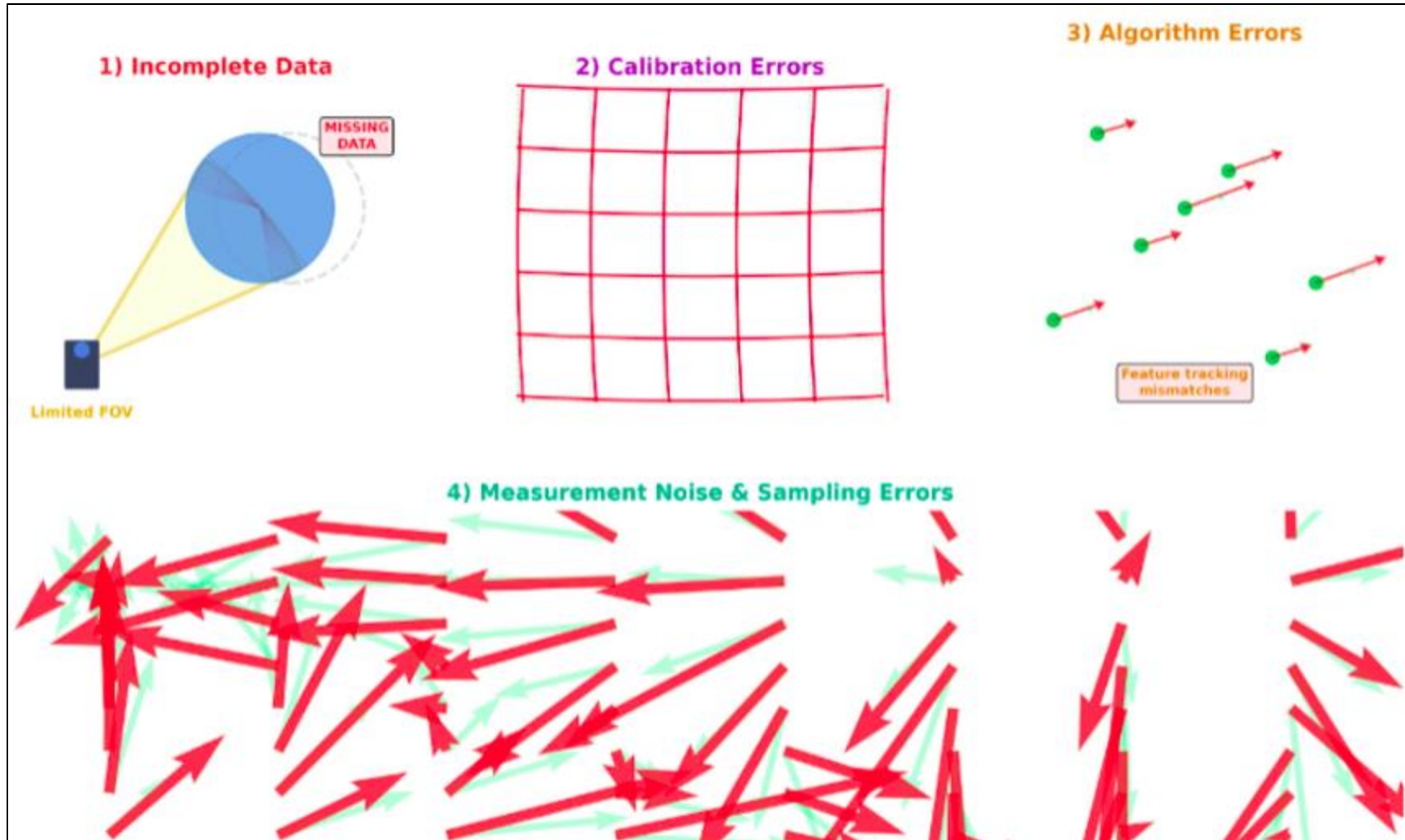
FUTURE WORK: THEORY & SIMULATION



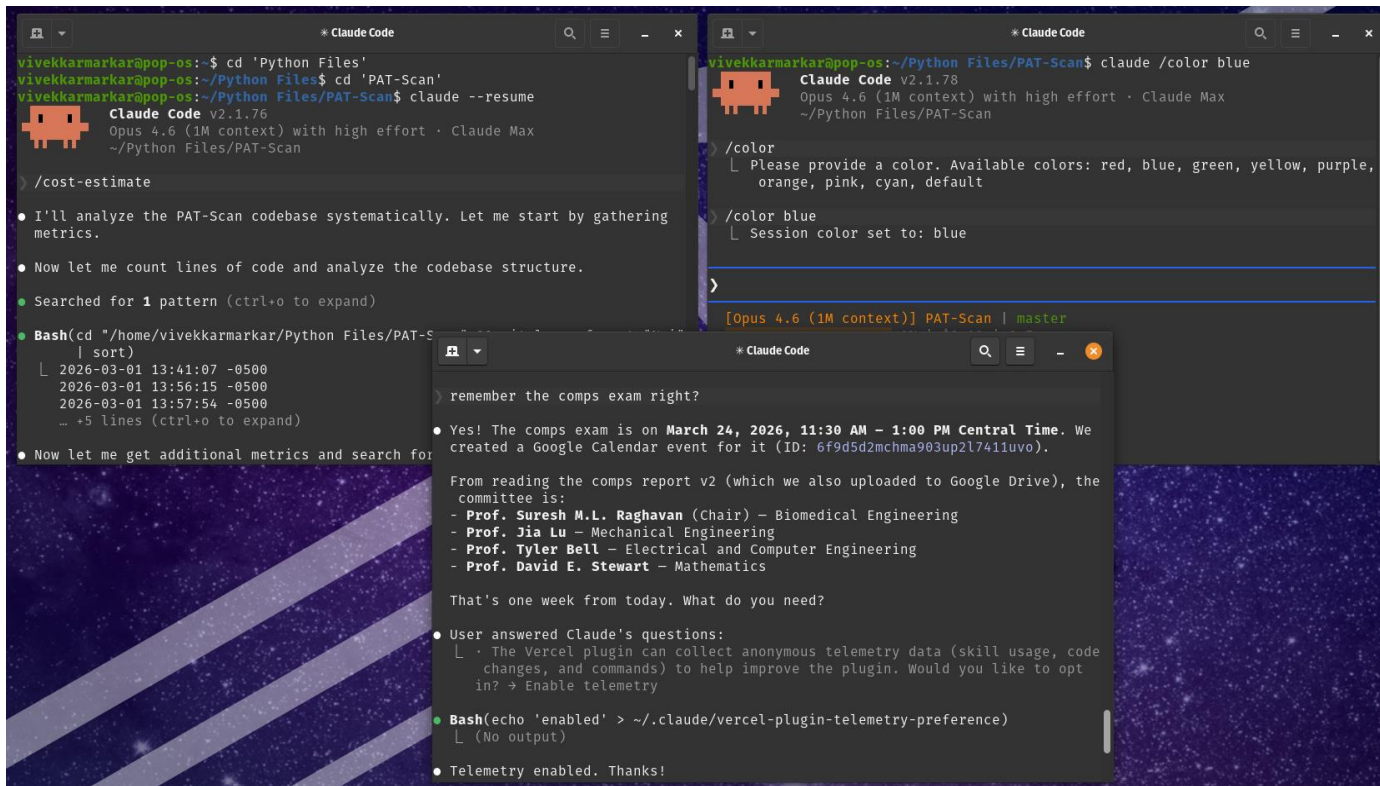
What is better:
Statics, Vibrations,
or a hybrid?



FUTURE WORK: EXPERIMENTAL CONSIDERATIONS



ACKNOWLEDGEMENTS



The image displays three overlapping screenshots of the Claude Code application interface. The background is a dark, starry space theme. The top-left screenshot shows the initial setup where the user navigates to the 'PAT-Scan' directory and runs the 'claude --resume' command. The top-right screenshot shows the user issuing the command 'claude /color blue', which sets the session color to blue. The bottom screenshot is a larger, more detailed view of the chat interface. It shows a conversation where the user asks 'remember the comps exam right?'. The assistant responds with a detailed answer, including the exam date (March 24, 2026, 11:30 AM - 1:00 PM Central Time) and a list of committee members: Prof. Suresh M.L. Raghavan (Chair), Prof. Jia Lu, Prof. Tyler Bell, and Prof. David E. Stewart. Below this, the assistant asks if the user needs anything else. The user then asks about the Vercel plugin's telemetry data collection, and the assistant explains it and offers to enable it. The user responds with 'enabled', and the assistant confirms that telemetry is now enabled.

```
vivekkarmarkar@pop-os:~$ cd 'Python Files'
vivekkarmarkar@pop-os:~/Python Files$ cd 'PAT-Scan'
vivekkarmarkar@pop-os:~/Python Files/PAT-Scan$ claude --resume

Claude Code v2.1.76
Opus 4.6 (1M context) with high effort · Claude Max
~/Python Files/PAT-Scan

> /cost-estimate

• I'll analyze the PAT-Scan codebase systematically. Let me start by gathering metrics.
• Now let me count lines of code and analyze the codebase structure.
• Searched for 1 pattern (ctrl+o to expand)
• Bash(cd "/home/vivekkarmarkar/Python Files/PAT-Scan" | sort)
  | 2026-03-01 13:41:07 -0500
  | 2026-03-01 13:56:15 -0500
  | 2026-03-01 13:57:54 -0500
  | ... +5 lines (ctrl+o to expand)
• Now let me get additional metrics and search for
```

```
vivekkarmarkar@pop-os:~/Python Files/PAT-Scan$ claude /color blue

Claude Code v2.1.78
Opus 4.6 (1M context) with high effort · Claude Max
~/Python Files/PAT-Scan

> /color
  | Please provide a color. Available colors: red, blue, green, yellow, purple, orange, pink, cyan, default
> /color blue
  | Session color set to: blue

>

[Opus 4.6 (1M context)] PAT-Scan | master
```

```
> remember the comps exam right?

• Yes! The comps exam is on March 24, 2026, 11:30 AM – 1:00 PM Central Time. We created a Google Calendar event for it (ID: 6f9d5d2mchma903up2l741luvo).

From reading the comps report v2 (which we also uploaded to Google Drive), the committee is:
- Prof. Suresh M.L. Raghavan (Chair) – Biomedical Engineering
- Prof. Jia Lu – Mechanical Engineering
- Prof. Tyler Bell – Electrical and Computer Engineering
- Prof. David E. Stewart – Mathematics

That's one week from today. What do you need?

• User answered Claude's questions:
  | • The Vercel plugin can collect anonymous telemetry data (skill usage, code changes, and commands) to help improve the plugin. Would you like to opt in? → Enable telemetry
• Bash(echo 'enabled' > ~/.claude/vercel-plugin-telemetry-preference)
  | (No output)
• Telemetry enabled. Thanks!
```

